

Universal Standard Model Effective Field Theory beyond dimension-six

Jay Desai

*T. Corbett, **J. Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia, M. Martines, and P. Reimitz* [arXiv:2304.03305](#)

*T. Corbett, **J. Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia* [arXiv:2404.03720](#)

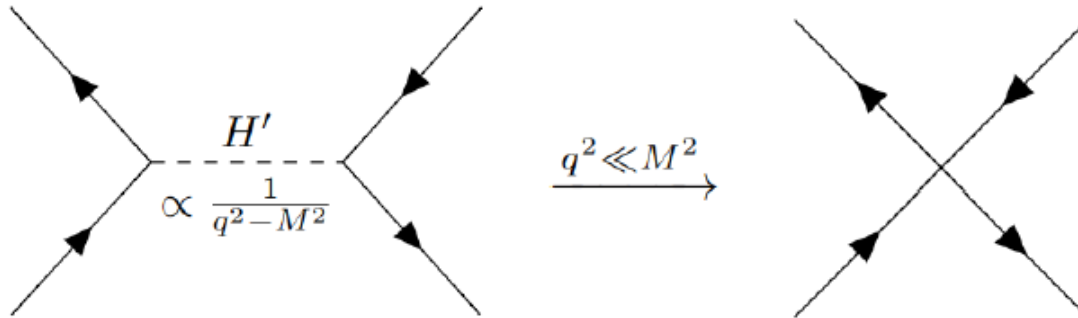
*T. Corbett, **J. Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia, M. Martines, and P. Reimitz* [arXiv:2503.19962](#)

Outline

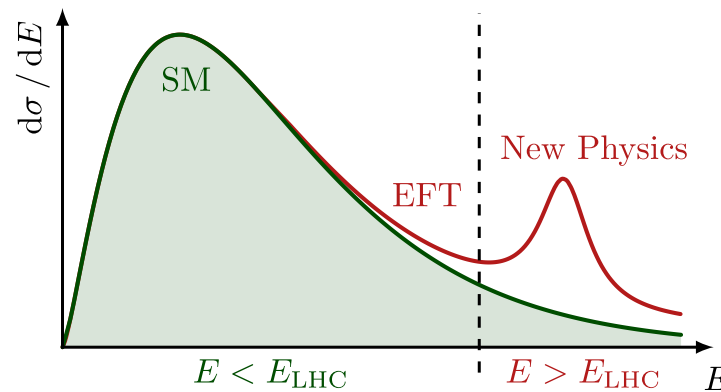
- Introduction
- USMEFT basis
- Universal effects in EWPO and DY
- Oblique Parameters
- Analysis
- Results

Standard Model Effective Field Theory

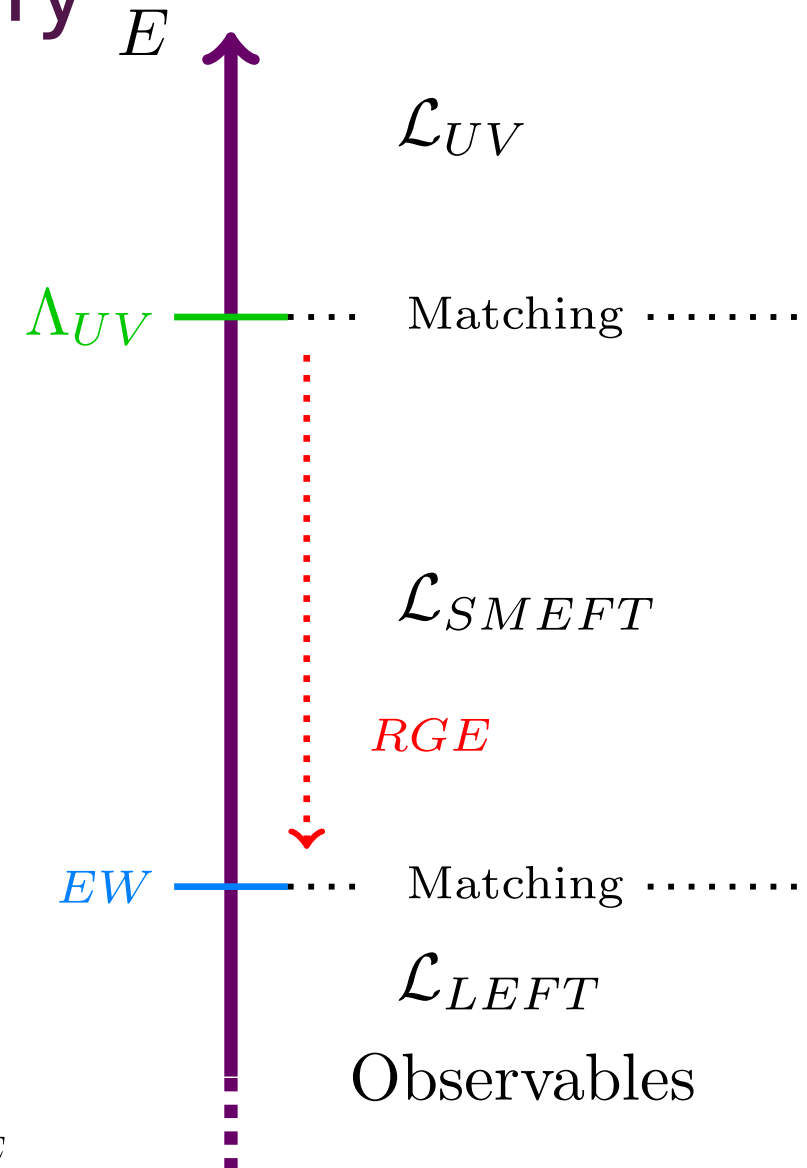
- Wilsonian perspective of a Quantum Field Theory gives us motivation for an Effective field theory approach



Indirect NP search



tikz.net/bsm_tails



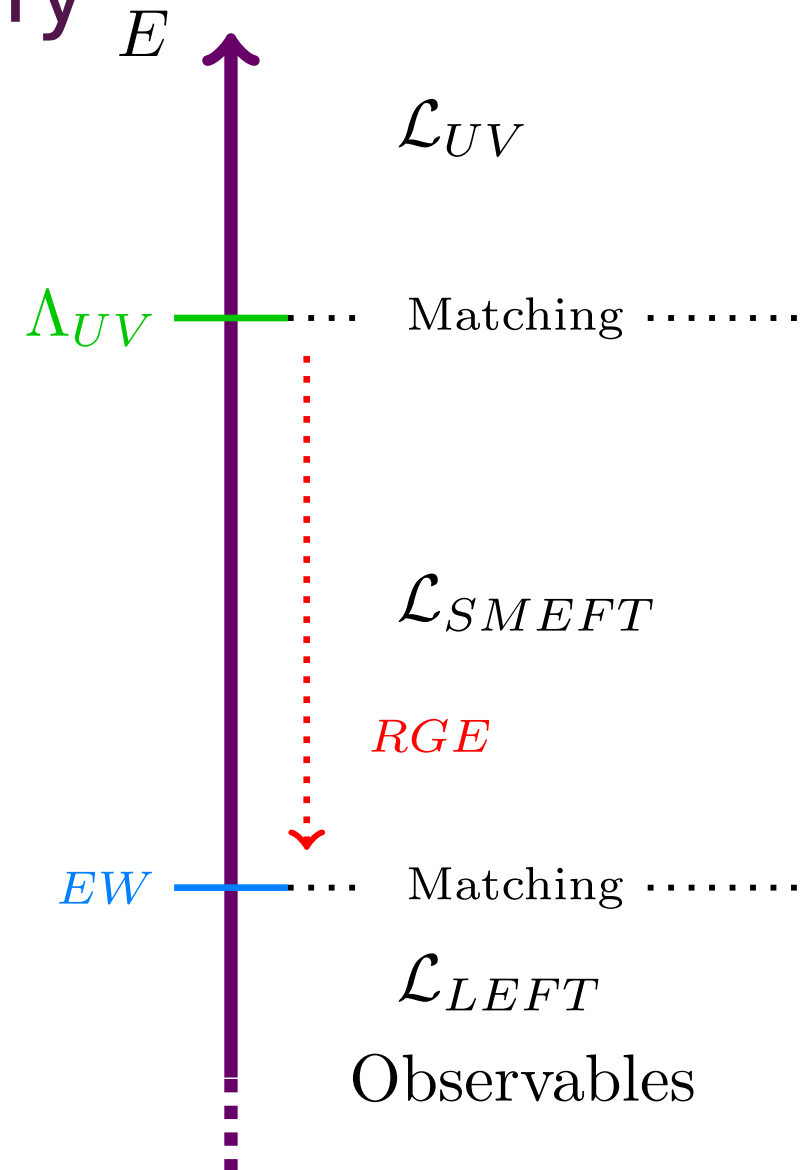
Standard Model Effective Field Theory

- Bottom-up approach organizes the EFT in a perturbative series where higher dimensional operators are suppressed by Λ

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_{j>4} \sum_n \frac{c_n^{(j)}}{\Lambda^{j-4}} \mathcal{O}_n^{(j)} ,$$

$$\text{BSM effects} \sim (v_{EW} / \Lambda_{UV})^{d-4}$$

$\Lambda_{UV} \gtrsim v_{EW}$: we need to go beyond leading order



Observables in SMEFT

- Within the SMEFT, to consistently include the corrections to observables at $\mathcal{O}(\Lambda^{-4})$ requires

$$|M_{\text{SM}}|^2 + \mathcal{M}_{\text{SM}}^* \mathcal{M}^{(6)} + |\mathcal{M}^{(6)}|^2 + \mathcal{M}_{\text{SM}}^* \mathcal{M}^{(6,2)} + \mathcal{M}_{\text{SM}}^* \mathcal{M}^{(8)}$$

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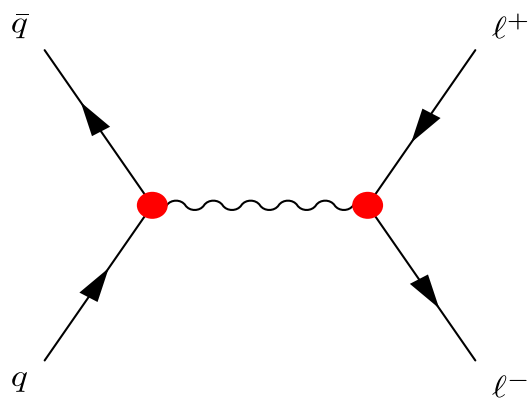
In some cases, dim-6 squared terms are important.

To consistently include next to leading order effects, we should include dimension-eight operators.

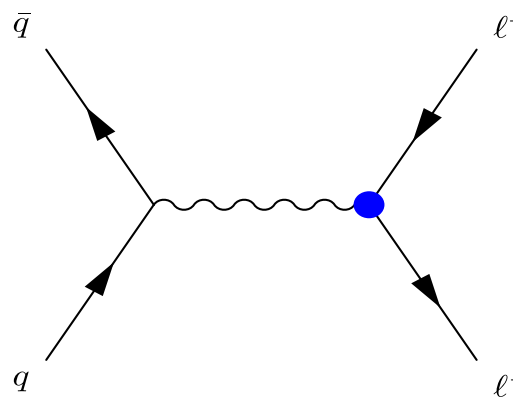
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● insertion of dim-6 operator

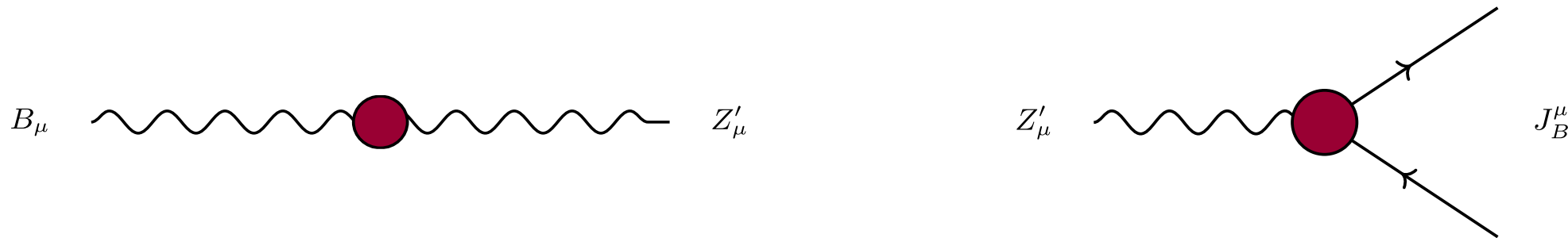


● insertion of dim-8 operator

Will the
dimension-eight fit
be robust?

Universal SMEFT (USMEFT)

- New Physics (NP) either dominantly couples to SM bosons or if it couples to fermions, it does so via SM currents.



Dim-6: J.D. Wells and Z. Zhang arXiv:1510.08462 [hep-ph] $\longrightarrow$ 16 operators

Dim-8: T. Corbett, **J. Desai**, O.J.P. Eboli, and M.C. Gonzalez-Garcia arXiv:2404.03720 [hep-ph]

$\longrightarrow$ 175 operators

Motivation

- In our previous work⁺, we saw that EWPO and EWDB data constrains the full parameter space of the relevant operators.

⁺ T. Corbett, **J. Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia, M. Martinez, and P. Reimitz
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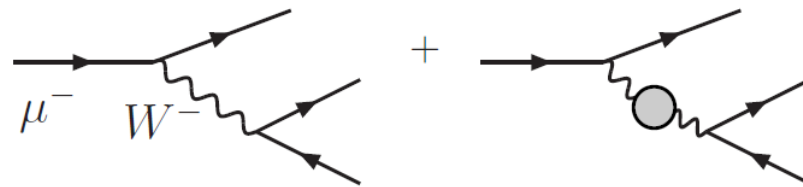
^{**}C.W. Murphy [arXiv:2005.00059 \[hep-ph\]](#)

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- What are the relevant fermionic operators that contribute to the $\Delta_{4F}^{(8)}$ parameter?



T. Corbett, J. **Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia [arXiv:2404.03720](#)

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- What are the relevant fermionic operators that contribute to the $\Delta_{4F}^{(8)}$ parameter?
- What are all the universal fermionic operators that are generated from trading the bosonic operators involving higher derivatives?

T. Corbett, J. **Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia [arXiv:2404.03720](#)

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Universal basis at dimension-eight

- We construct the complete operator basis for Universal SMEFT (USMEFT)* at dimension-eight, and we show that our basis of USMEFT is a subset of the general Murphy basis**.

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- We construct the complete operator basis for Universal SMEFT (USMEFT)* at dimension-eight, and we show that our basis of USMEFT is a subset of the general Murphy basis**.

- The building blocks of the operator basis for Universal theories are the Higgs field H , the SM field strengths ($X^{\mu\nu} \sim B^{\mu\nu}, W^{I,\mu\nu}, G^{A,\mu\nu}$) and covariant derivative D .

*T. Corbett, **J. Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia arXiv:2404.03720 [hep-ph]

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Universal basis at dimension-eight

- We construct the complete operator basis for Universal SMEFT (USMEFT)* at dimension-eight, and we show that our basis of USMEFT is a subset of the general Murphy basis**.

- Find the number of independent bosonic operators using packages BASISGEN and a modified version of ECO before applying EOM.



- Use Fierz relations and IBP relations to construct independent dimension-eight operators involving only SM bosons.

*T. Corbett, **J. Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia arXiv:2404.03720 [hep-ph]

**C.W. Murphy arXiv:2005.00059 [hep-ph]

C. Hays, A. Martin, V. Sanz, and J. Setford arXiv:1808.00442 [hep-ph]

Class of operators of D^2BH^4

$$x_1 = B^{\mu\nu}(D_\mu H^\dagger D_\nu H)(H^\dagger H) ,$$

$$x_2 = B^{\mu\nu}(D_\mu H^\dagger H)(H^\dagger D_\nu H) ,$$

$$x_3 = (D_\mu B^{\mu\nu})(D_\nu H^\dagger H)(H^\dagger H) ,$$

$$x_4 = (D_\mu B^{\mu\nu})(H^\dagger D_\nu H)(H^\dagger H) ,$$

$$x_5 = B^{\mu\nu}(D_\mu H^\dagger \tau^I D_\nu H)(H^\dagger \tau^I H) ,$$

$$x_6 = B^{\mu\nu}(D_\mu H^\dagger \tau^I H)(H^\dagger \tau^I D_\nu H) ,$$

$$x_7 = (D_\mu B^{\mu\nu})(D_\nu H^\dagger \tau^I H)(H^\dagger \tau^I H) ,$$

$$x_8 = (D_\mu B^{\mu\nu})(H^\dagger \tau^I D_\nu H)(H^\dagger \tau^I H) .$$

Linear SU(2) Fierz relations

$$x_5 = 2x_2 - x_1$$

$$x_6 = 2x_1 - x_2$$

$$x_7 = x_3$$

$$x_8 = x_4$$

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IBP relations

$$x_1 + x_2 - x_3 = 0$$

$$x_1 + x_2 + x_4 = 0$$

$$x_3 + x_4 = 0$$

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- We find 175 independent operators at dim8.
- 89 bosonic operators in SMEFT-8(Murphy) basis.
- 86 bosonic operators that involve higher derivatives.

- Find the number of independent bosonic operators using packages BASISGEN and a modified version of ECO before applying EOM.



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- Use equations of motion → fermionic operators in general SMEFT-8(Murphy) basis. → 86 independent combinations in Universal SMEFT-8 basis.



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In the general SMEFT (Murphy) basis.

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In the general SMEFT (Murphy) basis.

Use EOM,

$$D_\mu B^{\mu\nu} = -\frac{ig'}{2} H^\dagger \overleftrightarrow{D}^\nu H - J_B^\nu ,$$

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⇒ In the general SMEFT (Murphy) basis.

$$\Delta + i Q_{\psi^2 H^4 D}^{(1)} - i \frac{g'}{4} \left(Q_{\psi^2 H^5}^{(1)} + Q_{\psi^2 H^5}^{\dagger(1)} \right) ,$$

$$Q_{\psi^2 H^4 D}^{(1)} \equiv i J_B^\mu (H^\dagger \overleftrightarrow{D}_\mu H) (H^\dagger H)$$

$$Q_{\psi^2 H^5}^{(1)} \equiv (H^\dagger H)^2 (J_H H)$$

Use EOM,

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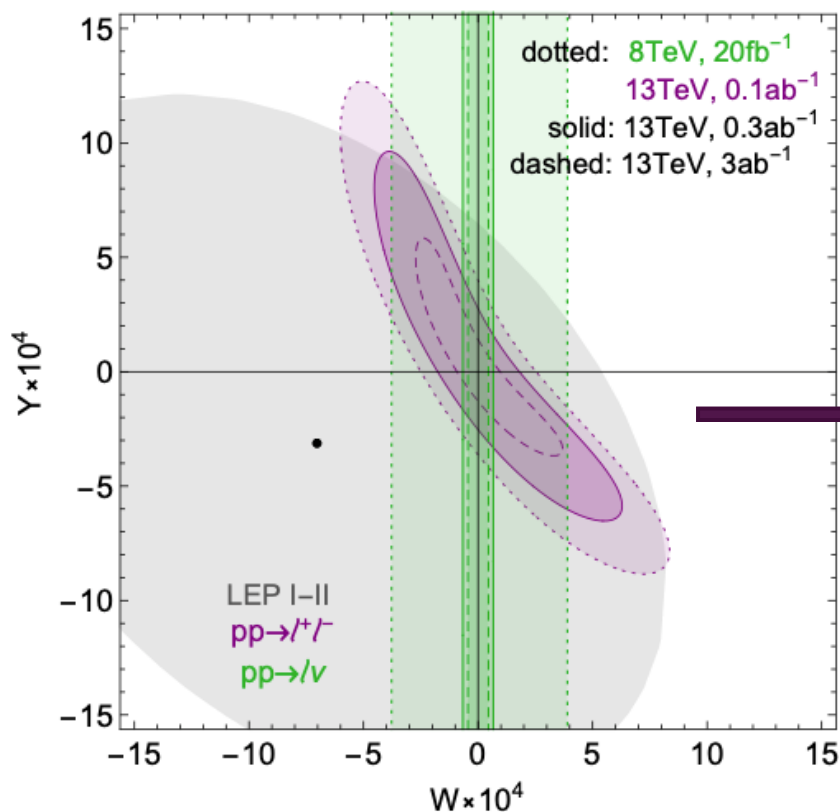
$$J_H^j = \sum_{ab} \left\{ y_{ab}^{u\dagger} (\bar{u}_a q_{bk}) \epsilon^{jk} + y_{ab}^d (\bar{q}_a^j d_b) + y_{ab}^e (\bar{l}_a^j e_b) \right\}$$

Drell Yan and EWPO beyond dimension-six

Energy Helps Accuracy

M. Farina et al. [1609.08157](#)

DY data can provide constraints that are competitive with those from LEP I-II data.



	universal form factor ($\mathcal{L}$)	contact operator ($\mathcal{L}'$)
W	$-\frac{W}{4m_W^2}(D_\rho W_{\mu\nu}^a)^2$	$-\frac{g_2^2 W}{2m_W^2}J_{L\mu}^a J_{La}^\mu$
Y	$-\frac{Y}{4m_W^2}(\partial_\rho B_{\mu\nu})^2$	$-\frac{g_1^2 Y}{2m_W^2}J_{Y\mu} J_Y^\mu$

Includes contributions from $d = 6$ and $(d = 6)^2$ terms.

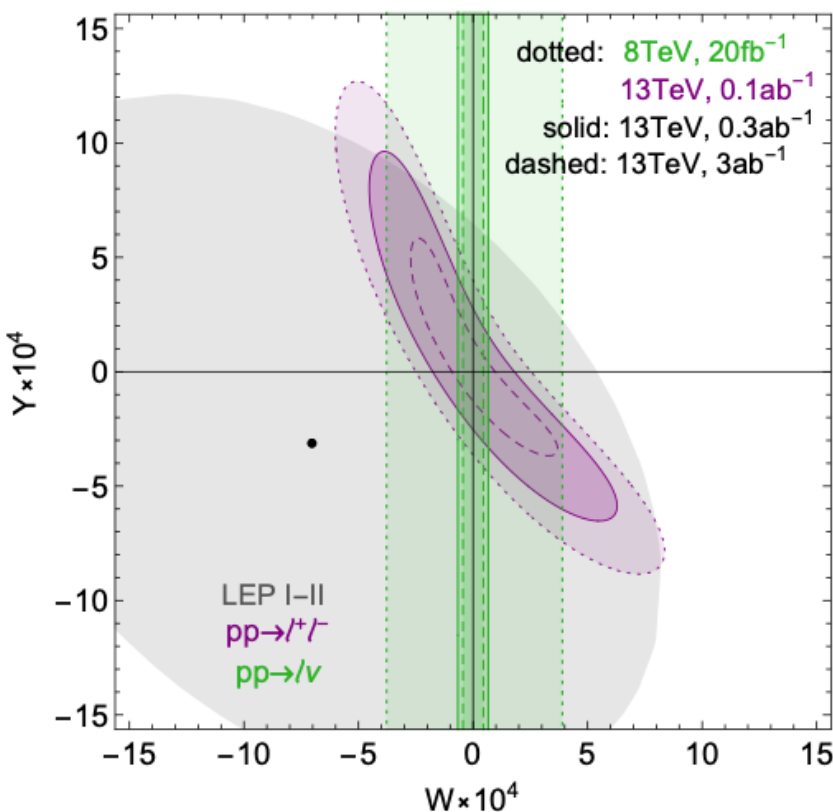
Will the same results hold when $d = 8$ terms are included?

M. Farina et al. [1609.08157](#)

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Dimension	$d = 6$			$d = 8$			
Operator classes	ψ^4	$\psi^2 H^2 D$	$\psi^2 XH$	$\psi^4 D^2$	$\psi^4 H^2$	$\psi^2 H^4 D$	$\psi^2 H^2 D^3$
Amplitude scaling	E^2/Λ^2	v^2/Λ^2	vE/Λ^2	E^4/Λ^4	$v^2 E^2/\Lambda^4$	v^4/Λ^4	$v^2 E^2/\Lambda^4$

L. Allwicher et al. 2207.10714

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Relevant dimension-six operators

Bosonic Basis		Coefficient	Rotated Basis		Coefficient
$Q_{\Phi,1}$	$(D_\mu H^\dagger H)(H^\dagger D^\mu H)$	$b_{\Phi,1}$	$Q_{\Phi,1}$	$(D_\mu H^\dagger H)(H^\dagger D^\mu H)$	$c_{\Phi,1}$
Q_{WW}	$H^\dagger W^I_{\mu\nu} W^{I,\mu\nu} H$	b_{WW}	Q_{WW}	$H^\dagger W^I_{\mu\nu} W^{I,\mu\nu} H$	c_{WW}
Q_{BB}	$H^\dagger B_{\mu\nu} B^{\mu\nu} H$	b_{BB}	Q_{BB}	$H^\dagger B_{\mu\nu} B^{\mu\nu} H$	c_{BB}
Q_{BW}	$H^\dagger B_{\mu\nu} \tau^I W^{I,\mu\nu} H$	b_{BW}	Q_{BW}	$H^\dagger B_{\mu\nu} \tau^I W^{I,\mu\nu} H$	c_{BW}
$\mathcal{R}_{2W}$	$-\frac{1}{2}(D^\nu W^I_{\mu\nu})^2$	r_{2W}	Q_{2JW}	$J^I_{W\mu} J^{I\mu}_W$	c_{2JW}
$\mathcal{R}_{2B}$	$-\frac{1}{2}(D^\nu B_{\mu\nu})^2$	r_{2B}	Q_{2JB}	$J_{B\mu} J^\mu_B$	c_{2JB}

$$J_B^\mu = g' \sum_{f \in \{q,l,u,d,e\}} \sum_a Y_f \bar{f}_a \gamma^\mu f_a ,$$

$$J_W^{I\mu} = \frac{g}{2} \sum_{f \in \{q,l\}} \sum_a \bar{f}_a \gamma^\mu \tau^I f_a$$

- We identify six independent bosonic operators contributing to the weak boson propagators.
- Using classical equations of motion, we trade two bosonic operators for operators involving fermions in the combination of SM currents.

Relevant dimension-eight operators

Bosonic Basis		Coeff	Rotated Basis		Coeff
$Q_{H^6}^{(2)}$	$(H^\dagger H)(H^\dagger \sigma^I H)(D_\mu H)^\dagger \sigma^I D^\mu H$	$b_{H^6}^{(2)}$	$Q_{H^6}^{(2)}$	$(H^\dagger H)(H^\dagger \sigma^I H)(D_\mu H)^\dagger \sigma^I D^\mu H$	$c_{H^6}^{(2)}$
$Q_{WBH^4}^{(1)}$	$(H^\dagger H)(H^\dagger \tau^I H)W_{\mu\nu}^I B^{\mu\nu}$	$b_{WBH^4}^{(1)}$	$Q_{WBH^4}^{(1)}$	$(H^\dagger H)(H^\dagger \tau^I H)W_{\mu\nu}^I B^{\mu\nu}$	$c_{WBH^4}^{(1)}$
$Q_{W^2H^4}^{(3)}$	$(H^\dagger \tau^I H)(H^\dagger \tau^J H)W_{\mu\nu}^I W^{J\mu\nu}$	$b_{W^2H^4}^{(3)}$	$Q_{W^2H^4}^{(3)}$	$(H^\dagger \tau^I H)(H^\dagger \tau^J H)W_{\mu\nu}^I W^{J\mu\nu}$	$c_{W^2H^4}^{(3)}$
$R_{B^2D^4}^{(1)}$	$D^\rho D^\alpha B_{\alpha\mu} D_\rho D_\beta B^{\beta\mu}$	$r_{B^2D^4}^{(1)}$	$Q_{\psi^4D^2}^{(2)}$	$D^\alpha J_B^\mu D_\alpha J_{B\mu}$	$c_{\psi^4D^2}^{(2)}$
$R_{W^2D^4}^{(1)}$	$D^\rho D^\alpha W_{\alpha\mu}^I D_\rho D^\beta W_\beta^{I,\mu}$	$r_{W^2D^4}^{(1)}$	$Q_{\psi^2H^2D^3}^{(1)}$	$i(D^\mu J_B^\nu + D^\nu J_B^\mu) \times (D_{(\mu} D_{\nu)} H^\dagger H - H^\dagger D_{(\mu} D_{\nu)} H)$	$c_{\psi^2H^2D^3}^{(1)}$
$R_{WH^4D^2}^{(1)}$	$i(D^\mu W_{\mu\nu}^I)(H^\dagger \overleftrightarrow{D}^{I\nu} H)(H^\dagger H)$	$r_{WH^4D^2}^{(1)}$	$Q_{\psi^4D^2}^{(3)}$	$D^\alpha J_W^{I\mu} D_\alpha J_{W\mu}^I$	$c_{\psi^4D^2}^{(3)}$
$R'_{WH^4D^2}{}^{(2)}$	$\epsilon^{IJK}(H^\dagger \tau^I H)D^\nu (H^\dagger \tau^J H)(D^\mu W_{\mu\nu}^K)$	$r'_{WH^4D^2}{}^{(2)}$	$Q_{\psi^2H^2D^3}^{(2)}$	$i(D^\mu J_W^{I\nu} + D^\nu J_W^{I\mu}) \times (D_{(\mu} D_{\nu)} H^\dagger \tau^I H - H^\dagger \tau^I D_{(\mu} D_{\nu)} H)$	$c_{\psi^2H^2D^3}^{(2)}$
$R_{BH^4D^2}^{(1)}$	$i(D_\alpha B^{\alpha\mu})(H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger H)$	$r_{BH^4D^2}^{(1)}$	$Q_{\psi^2H^4D}^{(1)}$	$iJ_B^\mu (H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger H)$	$c_{\psi^2H^4D}^{(1)}$
$R_{B^2H^2D^2}^{(9)}$	$(D^\mu B_{\mu\alpha})(D_\nu B^{\nu\alpha})(H^\dagger H)$	$r_{B^2H^2D^2}^{(9)}$	$Q_{\psi^2H^4D}^{(2)}$	$iJ_W^{I\mu} \left[(H^\dagger \overleftrightarrow{D}_\mu^I H)(H^\dagger H) + (H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger \tau^I H) \right]$	$c_{\psi^2H^4D}^{(2)}$
$R_{W^2H^2D^2}^{(9)}$	$(D^\mu W_{\mu\alpha}^I)(D_\nu W^{I,\nu\alpha})(H^\dagger H)$	$r_{W^2H^2D^2}^{(9)}$	$Q_{\psi^2H^4D}^{(4)}$	$\epsilon^{IJK} J_W^{I\mu} (H^\dagger \tau_\mu^J H) D_\mu (H^\dagger \tau^K H)$	$c_{\psi^2H^4D}^{(4)}$
$R_{BW H^2D^2}^{(13)}$	$(D^\mu B_{\mu\alpha})(D_\nu W^{I,\nu\alpha})(H^\dagger \tau^I H)$	$r_{BW H^2D^2}^{(13)}$	$Q_{\psi^4H^2}^{(4)}$	$J_B^\mu J_{B\mu}(H^\dagger H)$	$c_{\psi^4H^2}^{(4)}$
			$Q_{\psi^4H^2}^{(5)}$	$J_W^{I\mu} J_{W\mu}^I (H^\dagger H)$	$c_{\psi^4H^2}^{(5)}$
			$Q_{\psi^4H^2}^{(7)}$	$J_W^{I\mu} J_{B\mu}(H^\dagger \tau^I H)$	$c_{\psi^4H^2}^{(7)}$

$$c_{\psi^2H^2D^3}^{(1)} = -\frac{g'}{2} c_{\psi^4D^2}^{(2)} \ , \quad c_{\psi^2H^2D^3}^{(2)} = -\frac{g}{2} c_{\psi^4D^2}^{(3)} \ .$$

Relevant dimension-eight operators

		Bosonic Basis	Coeff			Rotated Basis	Coeff
3	{	$Q_{H^6}^{(2)}$	$(H^\dagger H)(H^\dagger \sigma^I H)(D_\mu H)^\dagger \sigma^I D^\mu H$	$b_{H^6}^{(2)}$	$Q_{H^6}^{(2)}$	$(H^\dagger H)(H^\dagger \sigma^I H)(D_\mu H)^\dagger \sigma^I D^\mu H$	$c_{H^6}^{(2)}$
		$Q_{WBH^4}^{(1)}$	$(H^\dagger H)(H^\dagger \tau^I H)W_{\mu\nu}^I B^{\mu\nu}$	$b_{WBH^4}^{(1)}$	$Q_{WBH^4}^{(1)}$	$(H^\dagger H)(H^\dagger \tau^I H)W_{\mu\nu}^I B^{\mu\nu}$	$c_{WBH^4}^{(1)}$
		$Q_{W^2H^4}^{(3)}$	$(H^\dagger \tau^I H)(H^\dagger \tau^J H)W_{\mu\nu}^I W^{J\mu\nu}$	$b_{W^2H^4}^{(3)}$	$Q_{W^2H^4}^{(3)}$	$(H^\dagger \tau^I H)(H^\dagger \tau^J H)W_{\mu\nu}^I W^{J\mu\nu}$	$c_{W^2H^4}^{(3)}$
8	{	$R_{B^2D^4}^{(1)}$	$D^\rho D^\alpha B_{\alpha\mu} D_\rho D_\beta B^{\beta\mu}$	$r_{B^2D^4}^{(1)}$	$Q_{\psi^4D^2}^{(2)}$	$D^\alpha J_B^\mu D_\alpha J_{B\mu}$	$c_{\psi^4D^2}^{(2)}$
					$Q_{\psi^2H^2D^3}^{(1)}$	$i(D^\mu J_B^\nu + D^\nu J_B^\mu) \times (D_{(\mu} D_{\nu)} H^\dagger H - H^\dagger D_{(\mu} D_{\nu)} H)$	$c_{\psi^2H^2D^3}^{(1)}$
		$R_{W^2D^4}^{(1)}$	$D^\rho D^\alpha W_{\alpha\mu}^I D_\rho D^\beta W_\beta^{I,\mu}$	$r_{W^2D^4}^{(1)}$	$Q_{\psi^4D^2}^{(3)}$	$D^\alpha J_W^{I\mu} D_\alpha J_{W\mu}^I$	$c_{\psi^4D^2}^{(3)}$
					$Q_{\psi^2H^2D^3}^{(2)}$	$i(D^\mu J_W^{I\nu} + D^\nu J_W^{I\mu}) \times (D_{(\mu} D_{\nu)} H^\dagger \tau^I H - H^\dagger \tau^I D_{(\mu} D_{\nu)} H)$	$c_{\psi^2H^2D^3}^{(2)}$
		$R_{WH^4D^2}^{(1)}$	$i(D^\mu W_{\mu\nu}^I)(H^\dagger \overleftrightarrow{D}^{I\nu} H)(H^\dagger H)$	$r_{WH^4D^2}^{(1)}$	$Q_{\psi^2H^4D}^{(1)}$	$iJ_B^\mu (H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger H)$	$c_{\psi^2H^4D}^{(1)}$
		$R_{WH^4D^2}^{\prime(2)}$	$\epsilon^{IJK}(H^\dagger \tau^I H)D^\nu (H^\dagger \tau^J H)(D^\mu W_{\mu\nu}^K)$	$r_{WH^4D^2}^{\prime(2)}$	$Q_{\psi^2H^4D}^{(2)}$	$iJ_W^{I\mu} \left[(H^\dagger \overleftrightarrow{D}_\mu^I H)(H^\dagger H) + (H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger \tau^I H) \right]$	$c_{\psi^2H^4D}^{(2)}$
		$R_{BH^4D^2}^{(1)}$	$i(D_\alpha B^{\alpha\mu})(H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger H)$	$r_{BH^4D^2}^{(1)}$	$Q_{\psi^2H^4D}^{(4)}$	$\epsilon^{IJK} J_W^{I\mu} (H^\dagger \tau_\mu^J H) D_\mu (H^\dagger \tau^K H)$	$c_{\psi^2H^4D}^{(4)}$
		$R_{B^2H^2D^2}^{(9)}$	$(D^\mu B_{\mu\alpha})(D_\nu B^{\nu\alpha})(H^\dagger H)$	$r_{B^2H^2D^2}^{(9)}$	$Q_{\psi^4H^2}^{(4)}$	$J_B^\mu J_{B\mu}(H^\dagger H)$	$c_{\psi^4H^2}^{(4)}$
			$Q_{\psi^4H^2}^{(5)}$	$J_W^{I\mu} J_{W\mu}^I (H^\dagger H)$	$c_{\psi^4H^2}^{(5)}$		
			$Q_{\psi^4H^2}^{(7)}$	$J_W^{I\mu} J_{B\mu}(H^\dagger \tau^I H)$	$c_{\psi^4H^2}^{(7)}$		

$$c_{\psi^2H^2D^3}^{(1)} = -\frac{g'}{2} c_{\psi^4D^2}^{(2)} \ , \quad c_{\psi^2H^2D^3}^{(2)} = -\frac{g}{2} c_{\psi^4D^2}^{(3)} \ .$$

Relevant dimension-eight operators

		Bosonic Basis	Coeff	Rotated Basis		Coeff	
3	{	$Q_{H^6}^{(2)}$	$(H^\dagger H)(H^\dagger \sigma^I H)(D_\mu H)^\dagger \sigma^I D^\mu H$	$b_{H^6}^{(2)}$	$Q_{H^6}^{(2)}$	$(H^\dagger H)(H^\dagger \sigma^I H)(D_\mu H)^\dagger \sigma^I D^\mu H$	$c_{H^6}^{(2)}$
		$Q_{WBH^4}^{(1)}$	$(H^\dagger H)(H^\dagger \tau^I H)W_{\mu\nu}^I B^{\mu\nu}$	$b_{WBH^4}^{(1)}$	$Q_{WBH^4}^{(1)}$	$(H^\dagger H)(H^\dagger \tau^I H)W_{\mu\nu}^I B^{\mu\nu}$	$c_{WBH^4}^{(1)}$
		$Q_{W^2H^4}^{(3)}$	$(H^\dagger \tau^I H)(H^\dagger \tau^J H)W_{\mu\nu}^I W^{J\mu\nu}$	$b_{W^2H^4}^{(3)}$	$Q_{W^2H^4}^{(3)}$	$(H^\dagger \tau^I H)(H^\dagger \tau^J H)W_{\mu\nu}^I W^{J\mu\nu}$	$c_{W^2H^4}^{(3)}$
8	{	$R_{B^2D^4}^{(1)}$	$D^\rho D^\alpha B_{\alpha\mu} D_\rho D_\beta B^{\beta\mu}$	$r_{B^2D^4}^{(1)}$	$Q_{\psi^4D^2}^{(2)}$	$D^\alpha J_B^\mu D_\alpha J_{B\mu}$	$c_{\psi^4D^2}^{(2)}$
		$R_{W^2D^4}^{(1)}$	$D^\rho D^\alpha W_{\alpha\mu}^I D_\rho D^\beta W_\beta^{I,\mu}$	$r_{W^2D^4}^{(1)}$	$Q_{\psi^2H^2D^3}^{(1)}$	$i(D^\mu J_B^\nu + D^\nu J_B^\mu) \times (D_{(\mu} D_{\nu)} H^\dagger H - H^\dagger D_{(\mu} D_{\nu)} H)$	$c_{\psi^2H^2D^3}^{(1)}$
		$R_{WH^4D^2}^{(1)}$	$i(D^\mu W_{\mu\nu}^I)(H^\dagger \overleftrightarrow{D}^{I\nu} H)(H^\dagger H)$	$r_{WH^4D^2}^{(1)}$	$Q_{\psi^4D^2}^{(3)}$	$D^\alpha J_W^{I\mu} D_\alpha J_{W\mu}^I$	$c_{\psi^4D^2}^{(3)}$
		$R_{WH^4D^2}^{\prime(2)}$	$\epsilon^{IJK}(H^\dagger \tau^I H)D^\nu (H^\dagger \tau^J H)(D^\mu W_{\mu\nu}^K)$	$r_{WH^4D^2}^{\prime(2)}$	$Q_{\psi^2H^2D^3}^{(2)}$	$i(D^\mu J_W^{I\nu} + D^\nu J_W^{I\mu}) \times (D_{(\mu} D_{\nu)} H^\dagger \tau^I H - H^\dagger \tau^I D_{(\mu} D_{\nu)} H)$	$c_{\psi^2H^2D^3}^{(2)}$
		$R_{BH^4D^2}^{(1)}$	$i(D_\alpha B^{\alpha\mu})(H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger H)$	$r_{BH^4D^2}^{(1)}$	$Q_{\psi^2H^4D}^{(1)}$	$iJ_B^\mu (H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger H)$	$c_{\psi^2H^4D}^{(1)}$
		$R_{B^2H^2D^2}^{(9)}$	$(D^\mu B_{\mu\alpha})(D_\nu B^{\nu\alpha})(H^\dagger H)$	$r_{B^2H^2D^2}^{(9)}$	$Q_{\psi^2H^4D}^{(2)}$	$iJ_W^{I\mu} \left[(H^\dagger \overleftrightarrow{D}_\mu^I H)(H^\dagger H) + (H^\dagger \overleftrightarrow{D}_\mu H)(H^\dagger \tau^I H) \right]$	$c_{\psi^2H^4D}^{(2)}$
		$R_{W^2H^2D^2}^{(9)}$	$(D^\mu W_{\mu\alpha}^I)(D_\nu W^{I,\nu\alpha})(H^\dagger H)$	$r_{W^2H^2D^2}^{(9)}$	$Q_{\psi^2H^4D}^{(4)}$	$\epsilon^{IJK} J_W^{I\mu} (H^\dagger \tau_\mu^J H) D_\mu (H^\dagger \tau^K H)$	$c_{\psi^2H^4D}^{(4)}$
		$R_{BWH^2D^2}^{(13)}$	$(D^\mu B_{\mu\alpha})(D_\nu W^{I,\nu\alpha})(H^\dagger \tau^I H)$	$r_{BWH^2D^2}^{(13)}$	$Q_{\psi^4H^2}^{(4)}$	$J_B^\mu J_{B\mu}(H^\dagger H)$	$c_{\psi^4H^2}^{(4)}$
			$Q_{\psi^4H^2}^{(5)}$	$J_W^{I\mu} J_{W\mu}^I (H^\dagger H)$	$c_{\psi^4H^2}^{(5)}$		
			$Q_{\psi^4H^2}^{(7)}$	$J_W^{I\mu} J_{B\mu}(H^\dagger \tau^I H)$	$c_{\psi^4H^2}^{(7)}$		

10

10

$$c_{\psi^2H^2D^3}^{(1)} = -\frac{g'}{2} c_{\psi^4D^2}^{(2)} \quad , \quad c_{\psi^2H^2D^3}^{(2)} = -\frac{g}{2} c_{\psi^4D^2}^{(3)} \quad .$$

Corrections to Drell-Yan amplitudes

- We have identified six dimension-six and 11 dimension-eight operators entering the Drell-Yan production in USMEFT.
- We adopt as input parameters

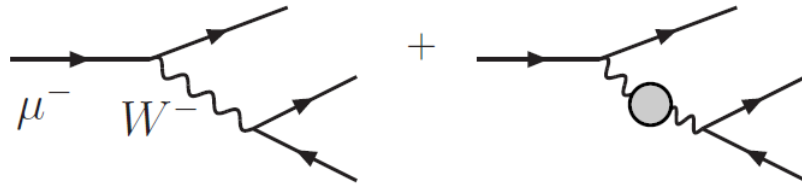
$$\{\hat{\alpha}_{\text{em}} , \hat{G}_F , \hat{M}_Z\}$$

$$\begin{aligned} \hat{e} &= \sqrt{4\pi\hat{\alpha}_{\text{em}}} , & \hat{c}^2 \hat{s}^2 &= \frac{\pi\hat{\alpha}_{\text{em}}}{\sqrt{2} \hat{G}_F \hat{M}_Z^2} , \\ \hat{v}^2 &= \frac{1}{\sqrt{2} \hat{G}_F} , \end{aligned}$$

- Higher dimensional operators will induce renormalization of the Standard Model couplings.

Finite renormalization of G_F

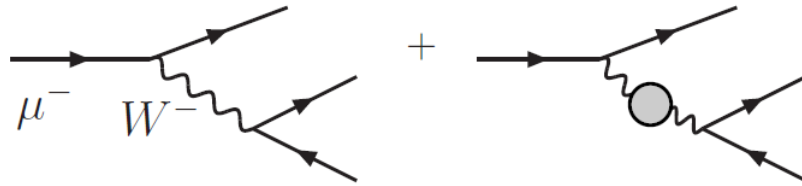
- The muon decay observable $\hat{G}_F$ is computed from the lifetime of the muon,



$$\left[2\langle H^\dagger H \rangle - \frac{1}{\sqrt{2}\hat{G}_F} \right]_{\text{fermionic}} \equiv \frac{\hat{v}^4}{\Lambda^2} \Delta_{4F}^{(6)} + \frac{\hat{v}^6}{\Lambda^4} \Delta_{4F}^{(8)} ,$$

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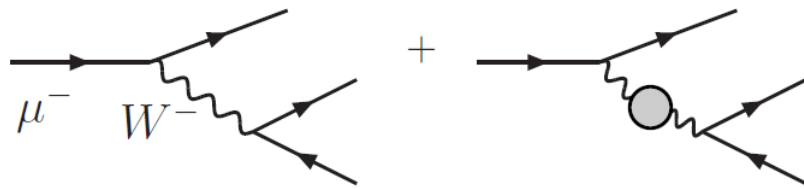
$$\left[2\langle H^\dagger H \rangle - \frac{1}{\sqrt{2}\hat{G}_F} \right]_{\text{fermionic}} \equiv \frac{\hat{v}^4}{\Lambda^2} \Delta_{4F}^{(6)} + \frac{\hat{v}^6}{\Lambda^4} \Delta_{4F}^{(8)} ,$$

$$\Delta_{4F}^{(6)} \sim Q_{2JW} = J_W^{I\mu} J_{W\mu}^I$$

$$J_W^{I\mu} = \frac{g}{2} \sum_{f \in \{q,l\}} \sum_a \bar{f}_a \gamma^\mu \tau^I f_a$$

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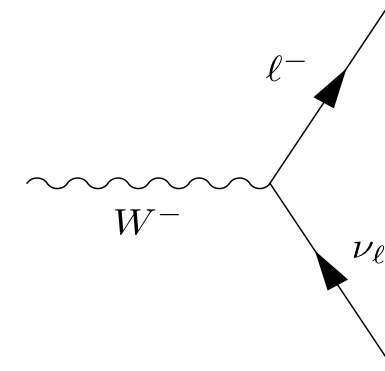
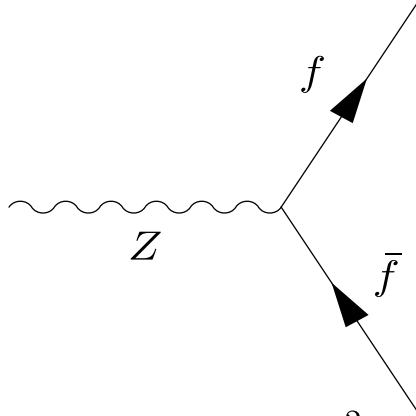
$$J_W^{I\mu} = \frac{g}{2} \sum_{f \in \{q,l\}} \sum_a \bar{f}_a \gamma^\mu \tau^I f_a$$

$$\Delta_{4F}^{(8)} \sim Q_{\psi^4 H^2}^{(5)} = J_W^{I\mu} J_{W\mu}^I (H^\dagger H)$$

$$Q_{\psi^2 H^4 D}^{(1)} = i J_B^\mu (H^\dagger \overleftrightarrow{D}_\mu H) (H^\dagger H)$$

$$Q_{\psi^2 H^4 D}^{(2)} = i J_W^{I\mu} \left[(H^\dagger \overleftrightarrow{D}_\mu^I H) (H^\dagger H) + (H^\dagger \overleftrightarrow{D}_\mu H) (H^\dagger \tau^I H) \right]$$

Corrections to Z and W couplings



$$\frac{\hat{e}}{\hat{s}\hat{c}} \left[\hat{g}_{L,R}^f \left(1 + \Delta\bar{g}_1 + \Delta g_1^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_1' \right) + Q^f \left(\Delta\bar{g}_2 + \Delta g_2^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_2' \right) \right], \quad \frac{1}{\sqrt{2}} \frac{\hat{e}}{\hat{s}} \left(1 + \Delta\bar{g}_W + \Delta g_W^\square + \frac{p^2}{\widehat{M}_W^2} \Delta g_W' \right),$$

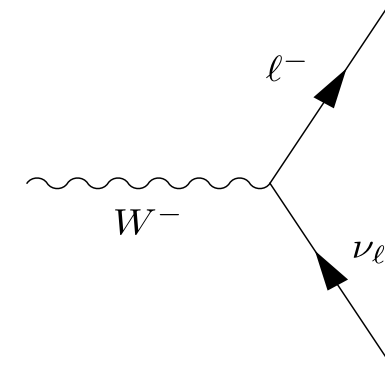
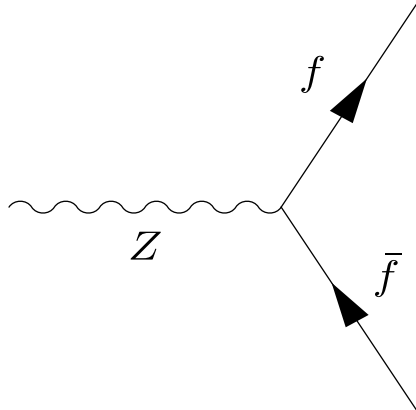
where $\hat{g}_L^f = T_3^f - \hat{s}^2 Q^f$, $\hat{g}_R^f = -\hat{s}^2 Q^f$

Dimension-six at
leading order

$$\begin{aligned} \Delta\bar{g}_1 &= -\frac{1}{4} [2\bar{\Delta}_{4F} + \bar{c}_{\Phi,1}] \frac{\hat{v}^2}{\Lambda^2}, \\ \Delta\bar{g}_2 &= -\frac{\hat{s}_2}{8\hat{c}_2} \left[\hat{s}_2 (2\bar{\Delta}_{4F} + \bar{c}_{\Phi,1}) + 4\bar{c}_{BW} \right] \frac{\hat{v}^2}{\Lambda^2}, \\ \Delta\bar{g}_W &= -\frac{1}{4\hat{c}_2} [2\hat{s}_2\bar{c}_{BW} + 2\hat{c}_2^2\bar{\Delta}_{4F} + \hat{c}_2^2\bar{c}_{\Phi,1}] \frac{\hat{v}^2}{\Lambda^2} - \frac{1}{2}\bar{c}_{W^2H^4}^{(3)} \frac{\hat{v}^4}{\Lambda^4}, \end{aligned}$$

with $\hat{c}_n = \cos(n\hat{\theta})$ and $\hat{s}_n = \sin(n\hat{\theta})$.

Corrections to Z and W couplings



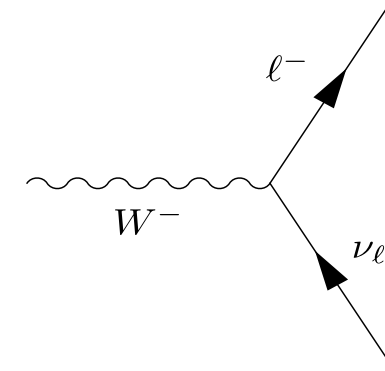
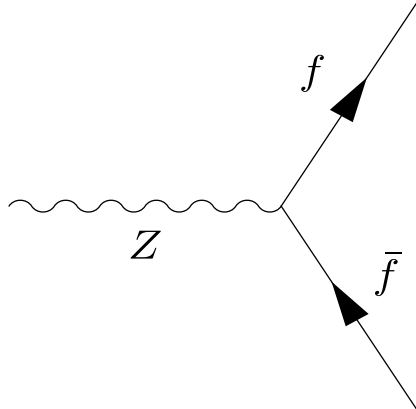
$$\frac{\hat{e}}{\hat{s}\hat{c}} \left[\hat{g}_{L,R}^f \left(1 + \Delta\bar{g}_1 + \Delta g_1^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_1' \right) + Q^f \left(\Delta\bar{g}_2 + \Delta g_2^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_2' \right) \right], \quad \frac{1}{\sqrt{2}} \frac{\hat{e}}{\hat{s}} \left(1 + \Delta\bar{g}_W + \Delta g_W^\square + \frac{p^2}{\widehat{M}_W^2} \Delta g_W' \right),$$

where $\hat{g}_L^f = T_3^f - \hat{s}^2 Q^f$, $\hat{g}_R^f = -\hat{s}^2 Q^f$

Includes additional
dimension-eight
operators that are only
resolvable with Higgs
observables

$$\begin{aligned} \bar{\Delta}_{4F} &= -\frac{\hat{e}^2}{2\hat{s}^2} c_{2JW} (1 - 2\frac{\hat{v}^2}{\Lambda^2} c_{WW}) - \frac{\hat{e}^2}{4\hat{s}^2} c_{\psi^4 H^2}^{(5)} \frac{\hat{v}^2}{\Lambda^2}, \\ \bar{c}_{BW} &= c_{BW} \left(1 - \frac{\hat{v}^2}{\Lambda^2} (c_{WW} + c_{BB}) \right) + \frac{1}{2} \left[c_{WBH^4}^{(1)} + \frac{\hat{e}}{2\hat{s}} c_{\psi^2 H^4 D}^{(1)} + \frac{\hat{e}}{\hat{c}} c_{\psi^2 H^4 D}^{(2)} \right] \frac{\hat{v}^2}{\Lambda^2}, \\ \bar{c}_{\Phi,1} &= c_{\Phi,1} + \left[c_{H^6}^{(2)} - \frac{\hat{e}}{\hat{c}} c_{\psi^2 H^4 D}^{(1)} - \frac{\hat{e}}{\hat{s}} (c_{\psi^2 H^4 D}^{(2)} - c_{\psi^2 H^4 D}^{(4)}) \right] \frac{\hat{v}^2}{\Lambda^2}, \\ \bar{c}_{W^2 H^4}^{(3)} &= c_{W^2 H^4}^{(3)} + \frac{\hat{e}}{2\hat{s}} \left(c_{\psi^2 H^4 D}^{(2)} - c_{\psi^2 H^4 D}^{(4)} \right). \end{aligned}$$

Corrections to Z and W couplings



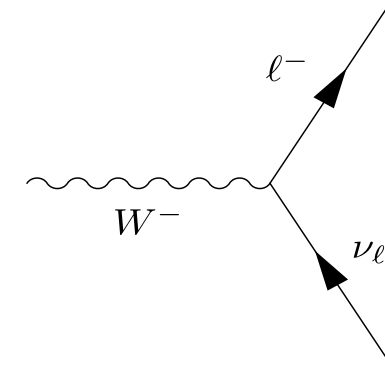
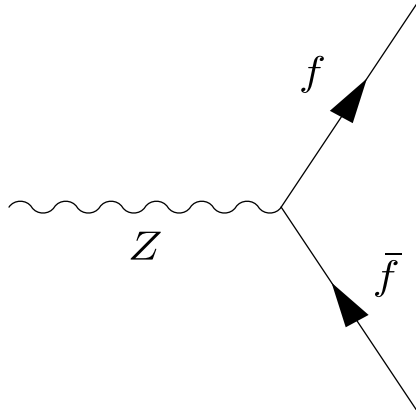
$$\frac{\hat{e}}{\hat{s}\hat{c}} \left[\hat{g}_{L,R}^f \left(1 + \Delta\bar{g}_1 + \Delta g_1^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_1' \right) + Q^f \left(\Delta\bar{g}_2 + \Delta g_2^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_2' \right) \right], \quad \frac{1}{\sqrt{2}} \frac{\hat{e}}{\hat{s}} \left(1 + \Delta\bar{g}_W + \Delta g_W^\square + \frac{p^2}{\widehat{M}_W^2} \Delta g_W' \right),$$

where $\hat{g}_L^f = T_3^f - \hat{s}^2 Q^f$, $\hat{g}_R^f = -\hat{s}^2 Q^f$

quadratic pieces $\longrightarrow \Delta g_1^\square = \frac{1}{32} \frac{1}{\hat{c}_2} \left[16\hat{s}^2 (\Delta_{4F})^2 + 3\hat{c}_2 (4(\Delta_{4F})^2 + (c_{\Phi,1})^2) + 4\Delta_{4F} c_{\Phi,1} + 16\hat{s}_2 \Delta_{4F} c_{BW} \right] \frac{\hat{v}^4}{\Lambda^4},$

momentum dependent $\longrightarrow \Delta g_1' = -\frac{\hat{e}^3}{4\hat{c}^2 \hat{s}^2} \left(\frac{1}{\hat{c}} c_{\psi^2 H^2 D^3}^{(1)} + \frac{1}{\hat{s}} c_{\psi^2 H^2 D^3}^{(2)} \right) \frac{\hat{v}^4}{\Lambda^4} = \frac{\hat{e}^4}{8\hat{c}^4 \hat{s}^4} \left(\hat{s}^2 c_{\psi^4 D^2}^{(2)} + \hat{c}^2 c_{\psi^4 D^2}^{(3)} \right) \frac{\hat{v}^4}{\Lambda^4},$

Corrections to Z and W couplings



$$\frac{\hat{e}}{\hat{s}\hat{c}} \left[\hat{g}_{L,R}^f \left(1 + \Delta\bar{g}_1 + \Delta g_1^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_1' \right) + Q^f \left(\Delta\bar{g}_2 + \Delta g_2^\square + \frac{p^2}{\widehat{M}_Z^2} \Delta g_2' \right) \right], \quad \frac{1}{\sqrt{2}} \frac{\hat{e}}{\hat{s}} \left(1 + \Delta\bar{g}_W + \Delta g_W^\square + \frac{p^2}{\widehat{M}_W^2} \Delta g_W' \right),$$

where $\hat{g}_L^f = T_3^f - \hat{s}^2 Q^f$, $\hat{g}_R^f = -\hat{s}^2 Q^f$

$$\Delta_{4F} = -\frac{\hat{e}^2}{2\hat{s}^2} c_{2JW}$$

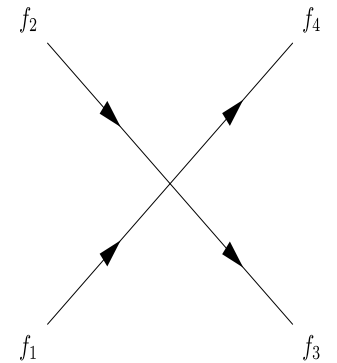
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momentum dependent $\longrightarrow \Delta g_1' = -\frac{\hat{e}^3}{4\hat{c}^2 \hat{s}^2} \left(\frac{1}{\hat{c}} c_{\psi^2 H^2 D^3}^{(1)} + \frac{1}{\hat{s}} c_{\psi^2 H^2 D^3}^{(2)} \right) \frac{\hat{v}^4}{\Lambda^4} = \frac{\hat{e}^4}{8\hat{c}^4 \hat{s}^4} \left(\hat{s}^2 c_{\psi^4 D^2}^{(2)} + \hat{c}^2 c_{\psi^4 D^2}^{(3)} \right) \frac{\hat{v}^4}{\Lambda^4},$

Four-fermion contact amplitudes

- In addition to the corrections to Z and W couplings, there are seven contact contributions to four-fermion amplitudes in the *rotated basis*.
 - two at dimension six: c_{2JB}, c_{2JW}
 - five at dimension eight: $c_{\psi^4 D^2}^{(2)}, c_{\psi^4 D^2}^{(3)}, c_{\psi^4 H^2}^{(4)}, c_{\psi^4 H^2}^{(5)}, c_{\psi^4 H^2}^{(7)}$.
- We find that the four-fermion contact amplitudes depend upon the two combinations

$$\begin{aligned}\bar{c}_{2JW} &\equiv c_{2JW} \left(1 - 2 \frac{\hat{v}^2}{\Lambda^2} c_{WW}\right) + c_{\psi^4 H^2}^{(5)} \frac{\hat{v}^2}{2\Lambda^2} = -\frac{2\hat{s}^2}{\hat{e}^2} \bar{\Delta}_{4F} , \\ \bar{c}_{2JB} &\equiv c_{2JB} \left(1 - 2 \frac{\hat{v}^2}{\Lambda^2} c_{BB}\right) + c_{\psi^4 H^2}^{(4)} \frac{\hat{v}^2}{2\Lambda^2} .\end{aligned}$$



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 - five at dimension eight: $c_{\psi^4 D^2}^{(2)}, c_{\psi^4 D^2}^{(3)}, c_{\psi^4 H^2}^{(4)}, c_{\psi^4 H^2}^{(5)}, c_{\psi^4 H^2}^{(7)}$.

4 effective dimension-six combinations

$$\bar{c}_{BW}, \bar{c}_{\phi,1}, \bar{\Delta}_{4F}, \bar{c}_{2JB}$$

4 effective dimension-eight combinations

$$\bar{c}_{W^2 H^4}^{(3)}, c_{\psi^4 H^2}^{(7)}, c_{\psi^4 D^2}^{(2)}, c_{\psi^4 D^2}^{(3)}$$

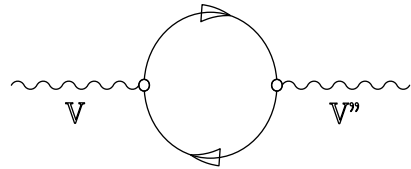
$$\bar{\Delta}_{4F} = -\frac{2\hat{s}^2}{\hat{e}^2} \bar{c}_{2JW}$$

Outline

- Introduction
- USMEFT basis
- Universal effects in EWPO and DY
- Oblique Parameters
- Analysis
- Results

Oblique Parameters

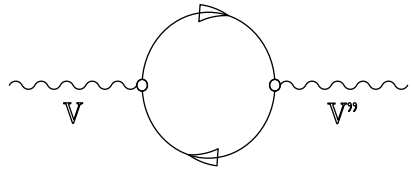
- In universal theories, for the *bosonic basis*, the effects of new physics can be parametrized in terms of the corrections to the gauge bosons self-energies. We expand the transverse vacuum polarization amplitudes as



$$\Pi_{VV''}(q^2) = \Pi_{VV''}(0) + \Pi'_{VV''}(0)q^2 + \frac{1}{2!}\Pi''_{VV''}(0)q^4 + \frac{1}{3!}\Pi'''_{VV''}(0)q^6 + \dots$$

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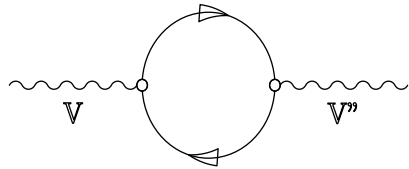
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To order q^4

$$\begin{aligned} \hat{S} &= -\frac{\hat{c}}{\hat{s}}\Pi_{3B}(0) , & \hat{T} &= \frac{1}{\widehat{M}_W^2} \left[\Pi_{WW}(0) - \Pi_{33}(0) \right] , \\ \hat{U} &= \Pi'_{33}(0) - \Pi'_{WW}(0) , & V &= \frac{\widehat{M}_W^2}{2} \left[\Pi''_{WW}(0) - \Pi''_{33}(0) \right] , \\ X &= -\frac{\widehat{M}_W^2}{2}\Pi''_{3B}(0) , & Y &= -\frac{\widehat{M}_W^2}{2}\Pi''_{BB}(0) , \\ W &= -\frac{\widehat{M}_W^2}{2}\Pi''_{33}(0) . \end{aligned}$$

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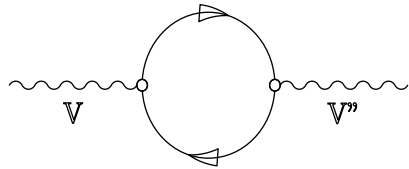
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Non-zero at
dimension-six in
SMEFT

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To order q^6

$$V' = \frac{\hat{M}_W^4}{6} \left[\Pi'''_{WW}(0) - \Pi'''_{33}(0) \right],$$

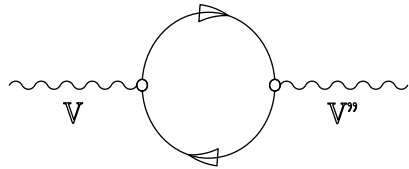
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Eight non-zero at dimension-eight in SMEFT

Modified propagators

- Incorporating the self-energy corrections one can write modified propagators for the NC and CC interactions, we extend the results from*

$$\Pi_{\gamma\gamma,\gamma Z,ZZ} \left(\begin{array}{c} \frac{1}{p^2 - \widehat{M}_Z^2} + \frac{2\Delta g_1^O}{p^2 - \widehat{M}_Z^2} - \frac{\left(\varepsilon_{ZZ} + \frac{2}{\hat{c}^2}\varepsilon'_{ZZ}\right)}{\widehat{M}_W^2} - \frac{\varepsilon'_{ZZ}}{\widehat{M}_W^4}p^2 \\ \frac{\Delta g_2^O}{\hat{s}\hat{c}(p^2 - \widehat{M}_Z^2)} - \frac{\left(\varepsilon_{Z\gamma} + \frac{1}{\hat{c}^2}\varepsilon'_{Z\gamma}\right)}{\widehat{M}_W^2} - \frac{\varepsilon'_{Z\gamma}}{\widehat{M}_W^4}p^2 \\ \frac{1}{p^2} - \frac{\varepsilon_{\gamma\gamma}}{\widehat{M}_W^2} - \frac{\varepsilon'_{\gamma\gamma}}{\widehat{M}_W^4}p^2 \end{array} \right),$$

$$\Pi_{WW} \frac{1}{p^2 - \widehat{M}_W^2} + \frac{2\Delta g_W^O}{p^2 - \widehat{M}_W^2} - \frac{\varepsilon_{WW} + 2\varepsilon'_{WW}}{\widehat{M}_W^2} - \frac{\varepsilon'_{WW}}{\widehat{M}_W^4}p^2$$

*M. Farina, G. Panico, D. Pappadopulo, J. T. Ruderman, R. Torre, and A. Wulzer et.al. arxiv:1609.08157 [hep-ph]

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Corrections to Z and W couplings

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Corrections to Z and W couplings

Mimics the effects from four-fermion interactions

*M. Farina, G. Panico, D. Pappadopulo, J. T. Ruderman, R. Torre, and A. Wulzer et.al. arxiv:1609.08157 [hep-ph]

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Analyses at different Orders

- $\mathcal{O}(\Lambda^{-4})$: including contributions at linear and quadratic order in dimension-six operator coefficients and at linear order in dimension-eight operator coefficients.
- $\mathcal{O}(\Lambda^{-2})$: including contributions only at linear order in dimension-six operator coefficients.
- Dim-6 + (Dim-6)²: including contributions at linear and quadratic order in dimension-six operator coefficients.
- Dim-6 + Dim-8: including contributions at linear order in both dimension-six and dimension-eight operator coefficients.

Analysis Framework

We identified six dimension-six and 11 dimension-eight operators entering the Drell-Yan production and EWPO in USMEFT.

4 effective dimension-six combinations

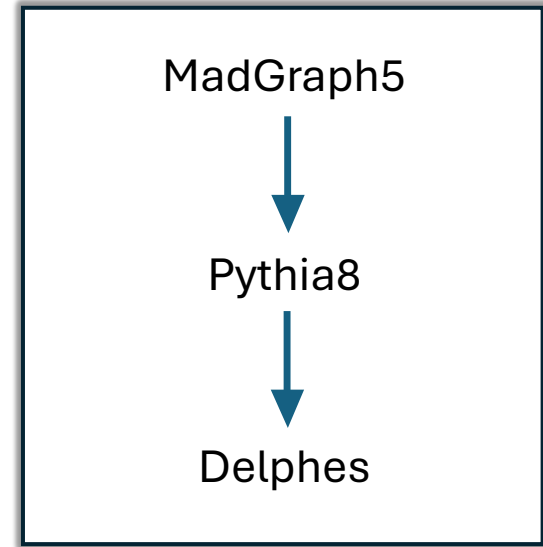
$$\bar{c}_{BW}, \bar{c}_{\phi,1}, \bar{\Delta}_{4F}, \bar{c}_{2JB}$$

4 effective dimension-eight combinations

$$\bar{c}_{W^2H^4}^{(3)}, c_{\psi^4H^2}^{(7)}, c_{\psi^4D^2}^{(2)}, c_{\psi^4D^2}^{(3)}$$

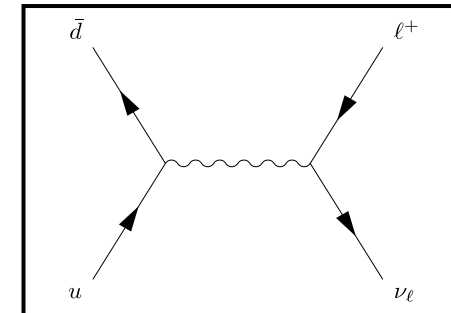
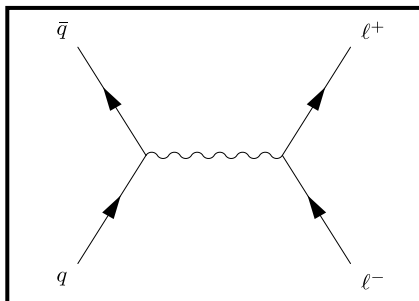
Data and simulation for Drell Yan

Channel	Distribution	# bins	ranges	data set	Int. Lum.
NC	$\frac{d^2\sigma}{dm_{\ell\ell}dy_{\ell\ell}}$	48	$116 \text{ GeV} \leq m_{\ell\ell} \leq 1.5 \text{ TeV}$ $0 \leq y_{\ell\ell} \leq 2.4$	ATLAS 8 TeV	20.3 fb^{-1}
NC	$\frac{dN_{\text{ev}}}{dm_{e^+e^-}}$	20	$250 \text{ GeV} \leq m_{e^+e^-} \leq 5 \text{ TeV}$	ATLAS 13 TeV	139 fb^{-1}
NC	$\frac{dN_{\text{ev}}}{dm_{\mu^+\mu^-}}$	20	$250 \text{ GeV} \leq m_{\mu^+\mu^-} \leq 5 \text{ TeV}$	ATLAS 13 TeV	139 fb^{-1}
NC	$\frac{dN_{\text{ev}}}{dm_{e^+e^-}}$	20	$300 \text{ GeV} \leq m_{e^+e^-} \leq 6 \text{ TeV}$	CMS 13 TeV	137 fb^{-1}
NC	$\frac{dN_{\text{ev}}}{dm_{\mu^+\mu^-}}$	20	$300 \text{ GeV} \leq m_{\mu^+\mu^-} \leq 7 \text{ TeV}$	CMS 13 TeV	137 fb^{-1}
CC	$\frac{d\sigma}{dm_T}$	20	$200 \text{ GeV} \leq m_{T,\ell\nu} \leq 5 \text{ TeV}$	ATLAS 13 TeV	140 fb^{-1}
CC	$\frac{dN}{dm_T}$	20	$440 \text{ GeV} \leq m_{T,e\nu} \leq 7 \text{ TeV}$	CMS 13 TeV	138 fb^{-1}
CC	$\frac{dN}{dm_T}$	20	$600 \text{ GeV} \leq m_{T,\mu\nu} \leq 7 \text{ TeV}$	CMS 13 TeV	138 fb^{-1}



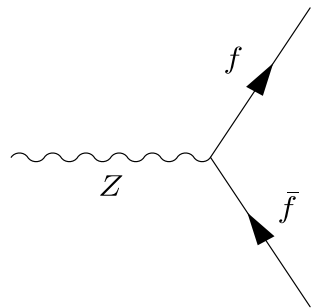
- The χ^2 function for DY depends on the eight Wilson coefficient combinations.

$$\chi_{\text{DY,NC}}^2(\bar{c}_{BW}, \bar{c}_{\Phi,1}, \bar{\Delta}_{4F}, \bar{c}_{2JB}, c_{\psi^4 D^2}^{(2)}, c_{\psi^4 D^2}^{(3)}, c_{\psi^4 H^2}^{(7)}) + \chi_{\text{DY,CC}}^2(\bar{c}_{BW}, \bar{c}_{\Phi,1}, \bar{\Delta}_{4F}, \bar{c}_{W^2 H^4}^{(3)}, c_{\psi^4 D^2}^{(3)}) .$$



Electro-Weak Precision Observables

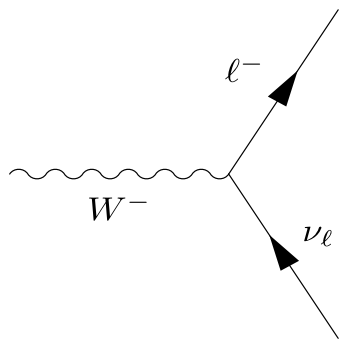
- We combine this with EWPO, where we include 12 Z-pole observables:



$\Gamma_Z, \sigma_h^0, \mathcal{A}_\ell(\tau^{\text{pol}}), R_\ell^0, \mathcal{A}_\ell(\text{SLD}), A_{\text{FB}}^{0,l}, R_c^0, R_b^0, \mathcal{A}_c, \mathcal{A}_b, A_{\text{FB}}^{0,c}, \text{ and } A_{\text{FB}}^{0,b}$

ALEPH, DELPHI, L3 and OPAL [arXiv:hep-ex/0509008](https://arxiv.org/abs/hep-ex/0509008) [hep-ex]

- We also include 2 W-pole observables:



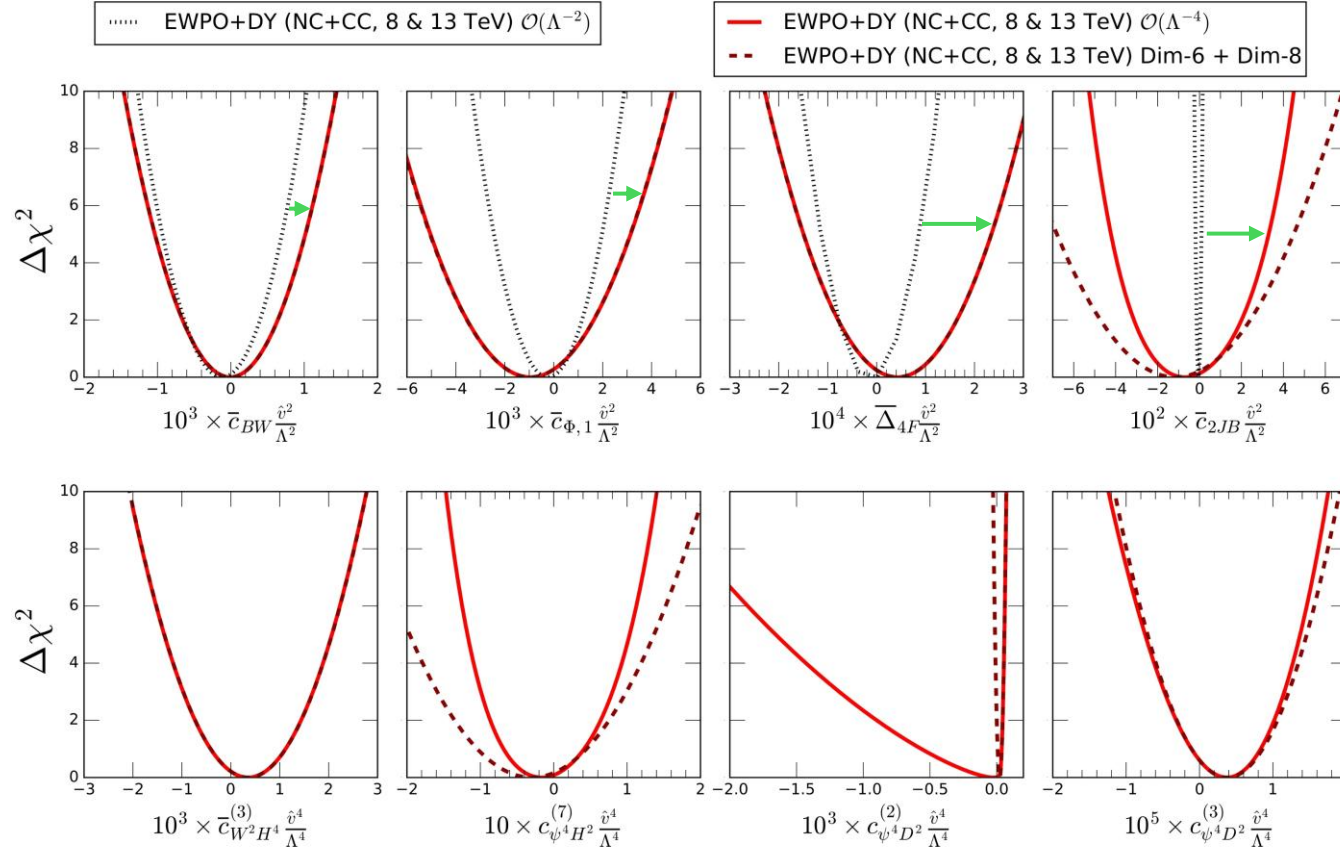
M_W and Γ_W

PDG [PhysRevD.110.030001]

$$\chi_{\text{EWPO}}^2(\bar{c}_{BW}, \bar{c}_{\Phi,1}, \bar{\Delta}_{4F}, \bar{c}_{W^2H^4}^{(3)}, c_{\psi^4D^2}^{(2)}, c_{\psi^4D^2}^{(3)}) .$$

Results

One dimensional projections

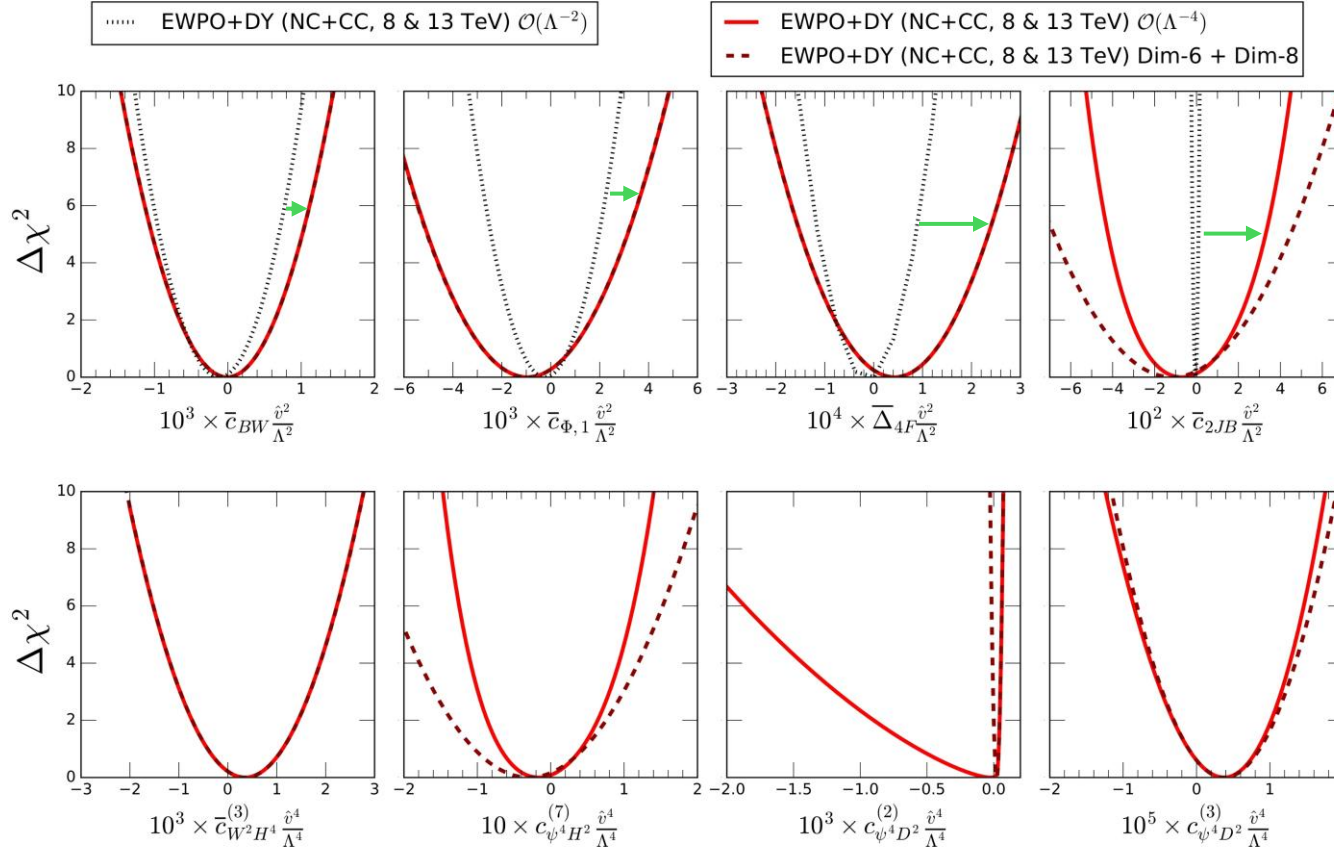


coupling	95% CL allowed range EWPO+DY			
	$\mathcal{O}(\Lambda^{-2})$	Dim-6+(Dim-6) ²	$\mathcal{O}(\Lambda^{-4})$	Dim-6+Dim-8
$\bar{c}_{BW} \frac{\hat{v}^2}{\Lambda^2}$	$[-8.5, 6.1] \times 10^{-4}$	$[-8.4, 6.1] \times 10^{-4}$	$[-9.3, 9.1] \times 10^{-4}$	$[-9.3, 9.1] \times 10^{-4}$
$\bar{c}_{\Phi,1} \frac{\hat{v}^2}{\Lambda^2}$	$[-2.2, 1.7] \times 10^{-3}$	$[-2.2, 1.7] \times 10^{-3}$	$[-4.6, 2.7] \times 10^{-3}$	$[-4.6, 2.7] \times 10^{-3}$
$\bar{\Delta}_{4F} \frac{\hat{v}^2}{\Lambda^2}$	$[-10, 7.6] \times 10^{-5}$	$[-10, 7.8] \times 10^{-5}$	$[-1.3, 2.1] \times 10^{-4}$	$[-1.3, 2.1] \times 10^{-4}$
$\bar{c}_{2JB} \frac{\hat{v}^2}{\Lambda^2}$	$[-3.6, 4.8] \times 10^{-4}$	$[-3.7, 5.0] \times 10^{-4}$	$[-10, 6.1] \times 10^{-4}$	$[-10, 6.1] \times 10^{-4}$
$\bar{c}_{2JB} \frac{\hat{v}^2}{\Lambda^2}$	$[-17, 7.9] \times 10^{-4}$	$[-21, 7.9] \times 10^{-4}$	$[-3.9, 2.9] \times 10^{-2}$	$[-6.2, 3.9] \times 10^{-2}$
$\bar{c}_{W^2 H^4}^{(3)} \frac{\hat{v}^4}{\Lambda^4}$			$[-1.4, 2.0] \times 10^{-3}$	$[-1.4, 2.0] \times 10^{-3}$
$c_{\psi^4 H^2}^{(7)} \frac{\hat{v}^4}{\Lambda^4}$			$[-1.1, 0.93] \times 10^{-1}$	$[-1.8, 1.1] \times 10^{-1}$
$c_{\psi^4 D^2}^{(2)} \frac{\hat{v}^4}{\Lambda^4}$			$[-14, 0.54] \times 10^{-4}$	$[-9.0, 54] \times 10^{-6}$
$c_{\psi^4 D^2}^{(3)} \frac{\hat{v}^4}{\Lambda^4}$			$[-6.1, 13] \times 10^{-6}$	$[-5.9, 13] \times 10^{-6}$

- The bounds on $\bar{c}_{BW}$, $\bar{c}_{\Phi,1}$, $\bar{\Delta}_{4F}$ become slightly weaker while bound on $\bar{c}_{2JB}$ becomes much weaker.
- The bounds are robust under the inclusion of dimension-six square contributions.

One dimensional projections

$$\bar{\Delta}_{4F} = -\frac{\hat{e}^2}{2\hat{s}^2}\bar{c}_{2JW}$$

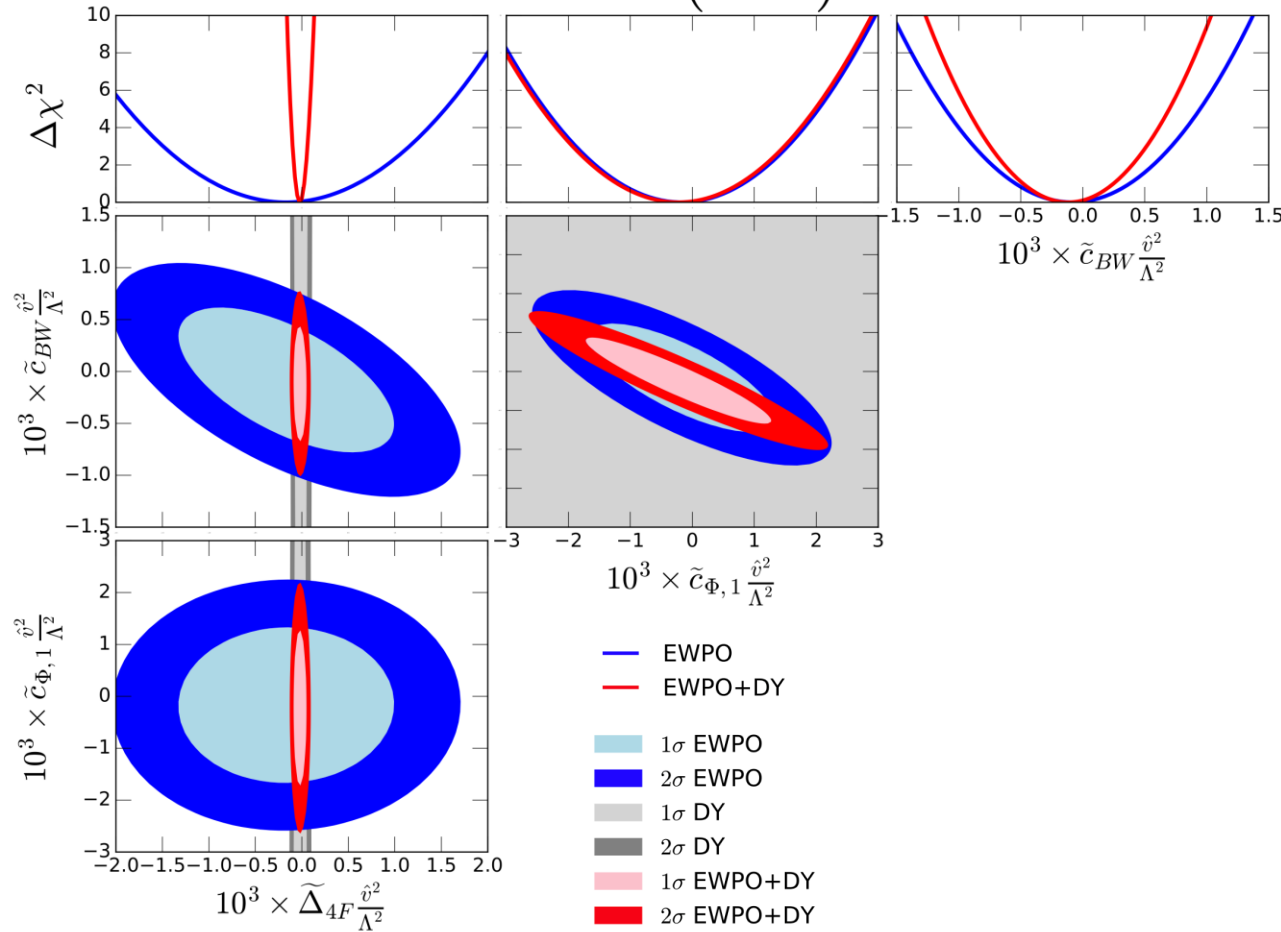


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EWPO vs. Drell-Yan

$\mathcal{O}(\Lambda^{-2})$



- The Z- and W- pole observables can constrain only four combinations of the coefficients.
- At dimension-six, only three coefficients

$$\tilde{\Delta}_{4F} = \bar{\Delta}_{4F} + \mathcal{O}(\Lambda^{-4})$$

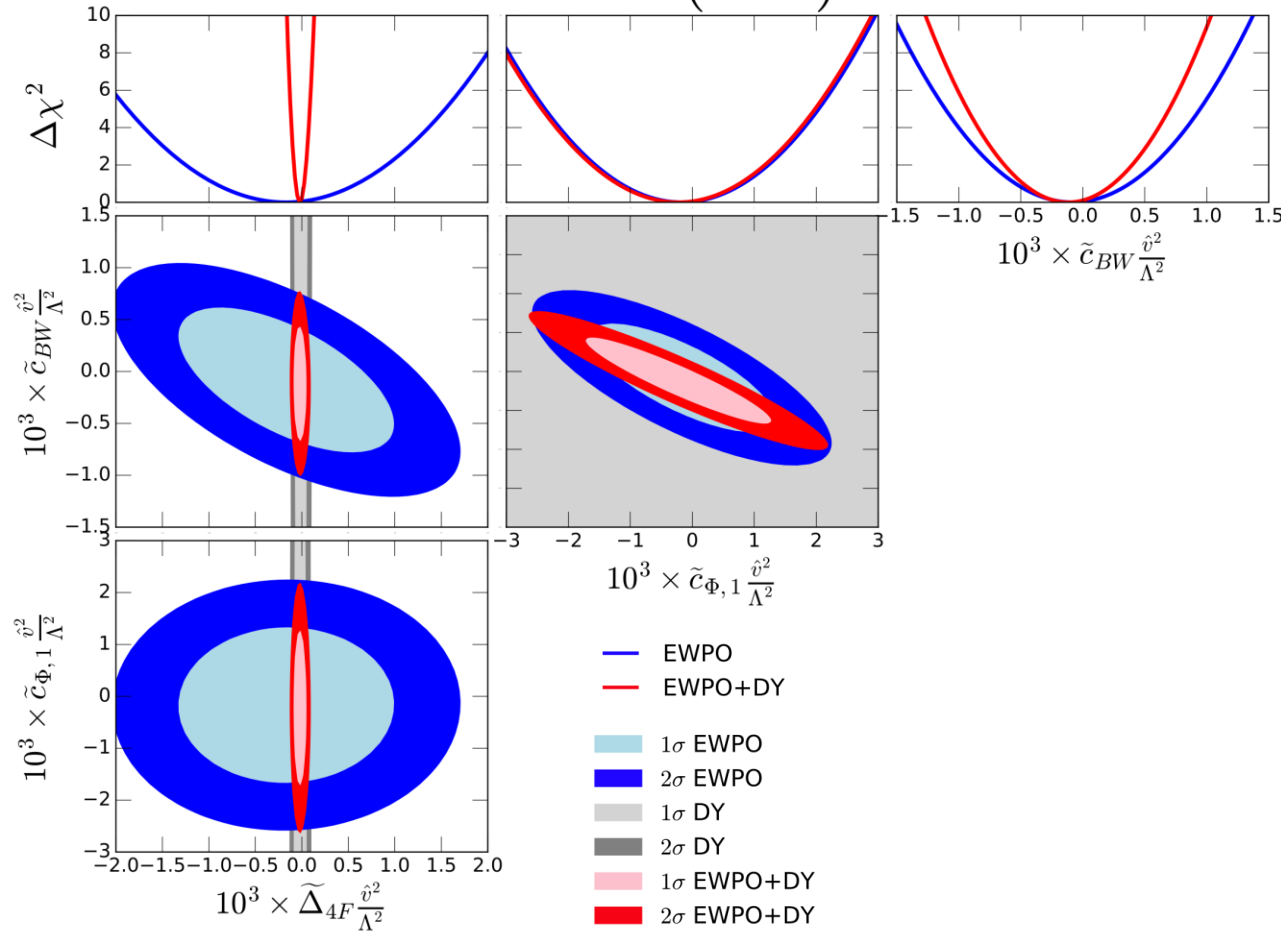
$$\tilde{c}_{BW} = \bar{c}_{BW} + \mathcal{O}(\Lambda^{-4})$$

$$\tilde{c}_{\phi,1} = \bar{c}_{\phi,1} + \mathcal{O}(\Lambda^{-4})$$

$$\begin{aligned} & \chi_{\text{EWPO}}^2(\bar{c}_{BW}, \bar{c}_{\Phi,1}, \bar{\Delta}_{4F}, \bar{c}_{W^2H^4}^{(3)}, c_{\psi^4D^2}^{(2)}, c_{\psi^4D^2}^{(3)}) \\ &= \chi_{\text{EWPO}}^2(\tilde{c}_{BW}, \tilde{c}_{\Phi,1}, \tilde{\Delta}_{4F}, \tilde{c}_{W^2H^4}^{(3)}) \end{aligned}$$

EWPO vs. Drell-Yan

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$$\tilde{c}_{BW} = \bar{c}_{BW} + \mathcal{O}(\Lambda^{-4})$$

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Corrections to Z and W couplings

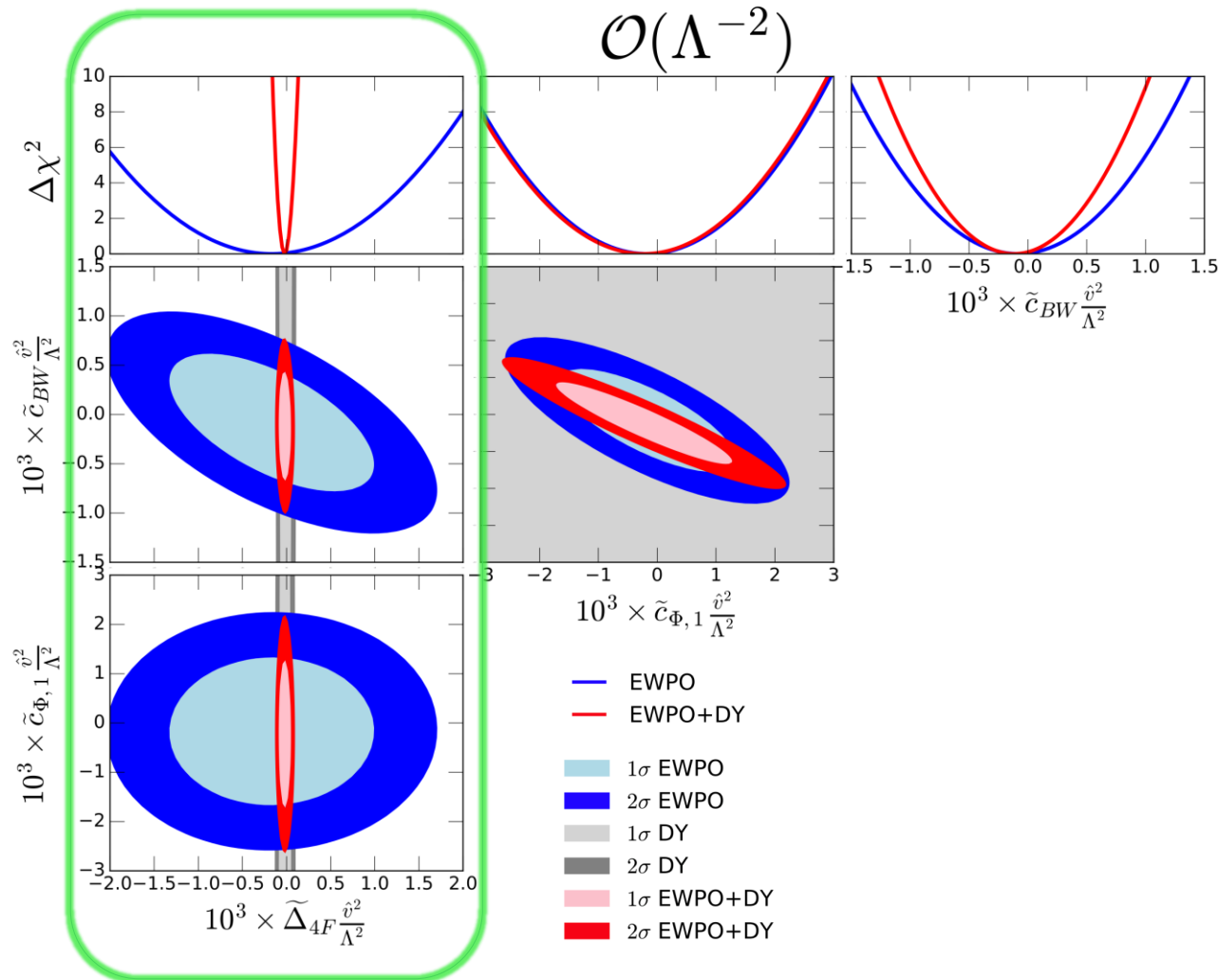
$$\Delta g_1^{Z \text{ pole}} = -\frac{1}{4} \left[2\tilde{\Delta}_{4F} + \tilde{c}_{\phi,1} \right] \frac{\hat{v}^2}{\Lambda^2},$$

$$\Delta g_2^{Z \text{ pole}} = -\frac{\hat{s}_2}{8\hat{c}_2} \left[\hat{s}_2 \left(2\tilde{\Delta}_{4F} + \tilde{c}_{\phi,1} \right) + 4\tilde{c}_{BW} \right] \frac{\hat{v}^2}{\Lambda^2},$$

$$\Delta g_W^{W \text{ pole}} = -\frac{1}{4\hat{c}_2} \left[2\hat{s}_2\tilde{c}_{BW} + 2\hat{c}^2\tilde{\Delta}_{4F} + \hat{c}^2\tilde{c}_{\phi,1} \right] \frac{\hat{v}^2}{\Lambda^2},$$

$$\frac{\Delta M_W}{\widehat{M}_W} = -\frac{1}{4\hat{c}_2} \frac{\hat{v}^2}{\Lambda^2} \left[2\hat{s}_2\tilde{c}_{BW} + 2\hat{s}_2^2\tilde{\Delta}_{4F} + \hat{c}^2\tilde{c}_{\phi,1} \right].$$

EWPO vs. Drell-Yan



- The Z- and W- pole observables can constrain only four combinations of the coefficients.
- At dimension-six, only three coefficients

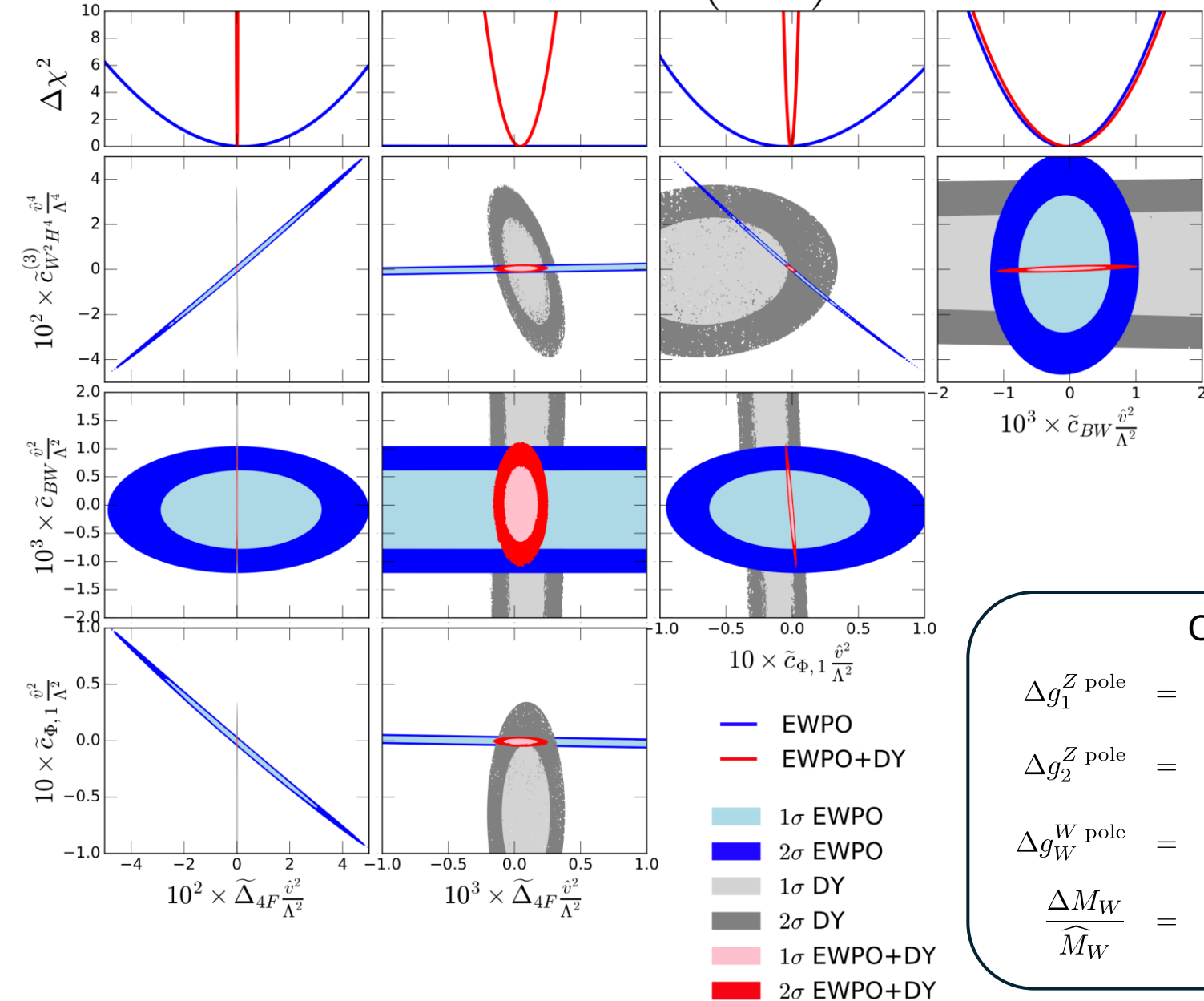
$$\tilde{\Delta}_{4F} = \bar{\Delta}_{4F} + \mathcal{O}(\Lambda^{-4})$$

$$\tilde{c}_{BW} = \bar{c}_{BW} + \mathcal{O}(\Lambda^{-4})$$

$$\tilde{c}_{\phi,1} = \bar{c}_{\phi,1} + \mathcal{O}(\Lambda^{-4})$$

Limits on Δ_{4F} are stronger when including DY data!

$\mathcal{O}(\Lambda^{-4})$



At dimension-eight U is generated.

$$\tilde{\Delta}_{4F} = \bar{\Delta}_{4F} - \frac{\hat{e}^4}{4\hat{s}^4} c_{\psi^4 D^2}^{(3)} \frac{\hat{v}^2}{\Lambda^2}$$

$$\tilde{c}_{BW} = \bar{c}_{BW} + \frac{\hat{e}^4}{8\hat{s}^3 \hat{c}^3} \left(c_{\psi^4 D^2}^{(2)} + c_{\psi^4 D^2}^{(3)} \right) \frac{\hat{v}^2}{\Lambda^2}$$

$$\tilde{c}_{\Phi,1} = \bar{c}_{\Phi,1} - \frac{\hat{e}^4}{2\hat{s}^2 \hat{c}^4} \left(c_{\psi^4 D^2}^{(2)} + \hat{c}^2 c_{\psi^4 D^2}^{(3)} \right) \frac{\hat{v}^2}{\Lambda^2}$$

$$\tilde{c}_{W^2 H^4}^{(3)} = \bar{c}_{W^2 H^4}^{(3)} + \frac{\hat{e}^4}{4\hat{s}^2 \hat{c}^2} c_{\psi^4 D^2}^{(3)}$$

Corrections to Z and W couplings

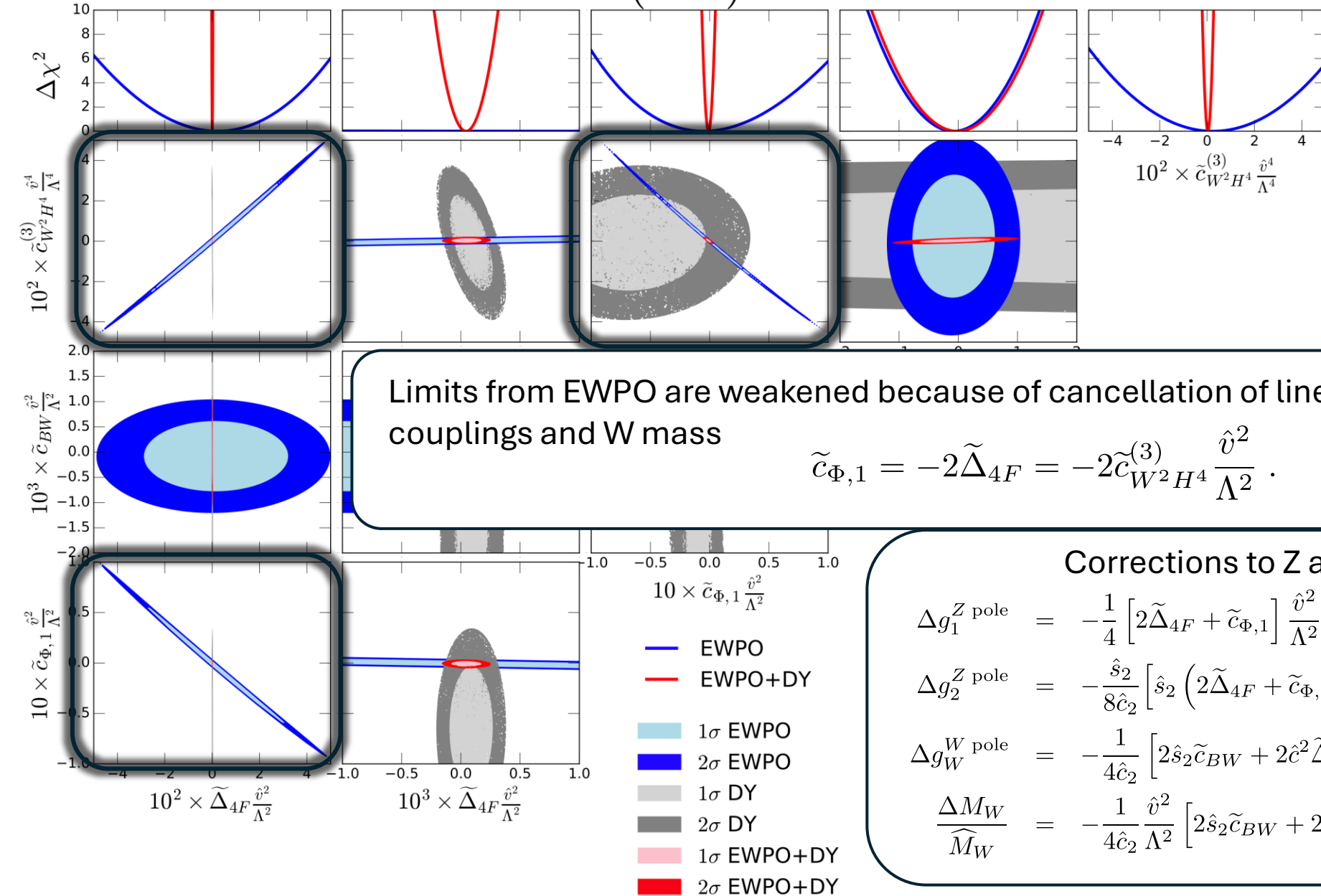
$$\Delta g_1^{Z \text{ pole}} = -\frac{1}{4} \left[2\tilde{\Delta}_{4F} + \tilde{c}_{\Phi,1} \right] \frac{\hat{v}^2}{\Lambda^2} + \dots,$$

$$\Delta g_2^{Z \text{ pole}} = -\frac{\hat{s}_2}{8\hat{c}_2} \left[\hat{s}_2 \left(2\tilde{\Delta}_{4F} + \tilde{c}_{\Phi,1} \right) + 4\tilde{c}_{BW} \right] \frac{\hat{v}^2}{\Lambda^2} + \dots,$$

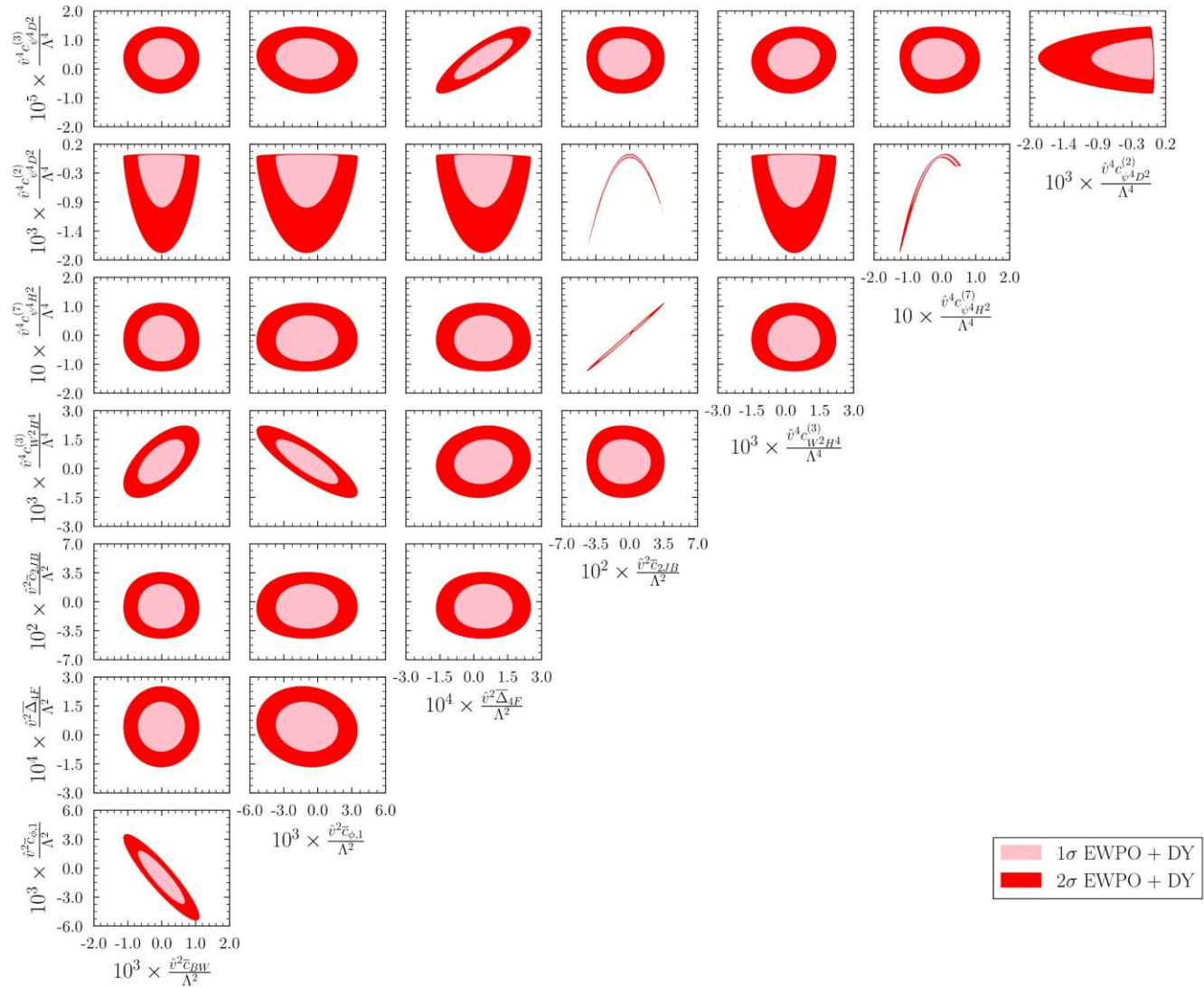
$$\Delta g_W^{W \text{ pole}} = -\frac{1}{4\hat{c}_2} \left[2\hat{s}_2 \tilde{c}_{BW} + 2\hat{c}^2 \tilde{\Delta}_{4F} + \hat{c}^2 \tilde{c}_{\Phi,1} \right] \frac{\hat{v}^2}{\Lambda^2} - \frac{1}{2} \tilde{c}_{W^2 H^4}^{(3)} \frac{\hat{v}^4}{\Lambda^4} + \dots,$$

$$\frac{\Delta M_W}{\widehat{M}_W} = -\frac{1}{4\hat{c}_2} \frac{\hat{v}^2}{\Lambda^2} \left[2\hat{s}_2 \tilde{c}_{BW} + 2\hat{s}^2 \tilde{\Delta}_{4F} + \hat{c}^2 \tilde{c}_{\Phi,1} \right] - \frac{1}{2} \tilde{c}_{W^2 H^4}^{(3)} \frac{\hat{v}^4}{\Lambda^4} + \dots$$

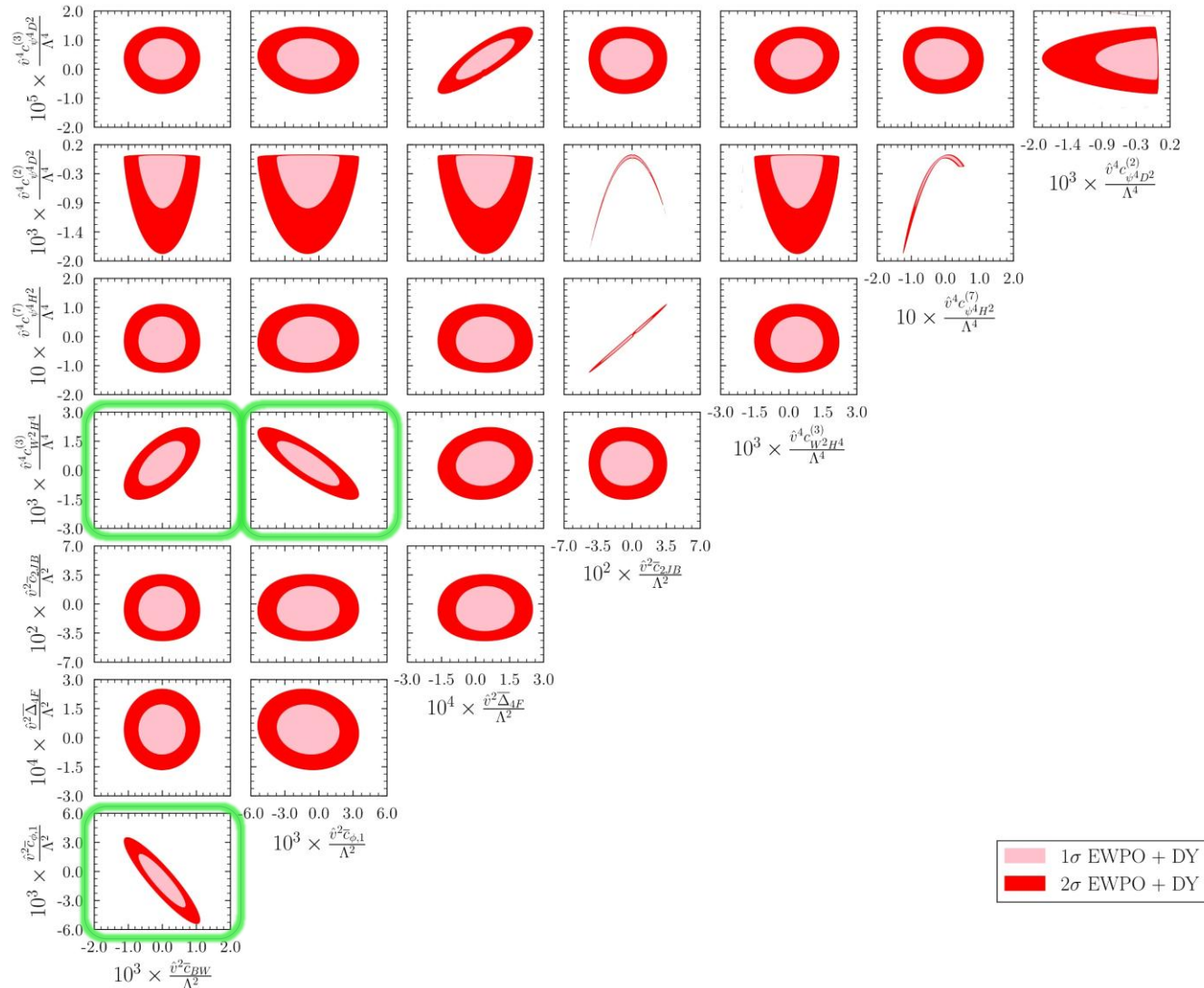
$$\mathcal{O}(\Lambda^{-4})$$



Two-dimensional projections

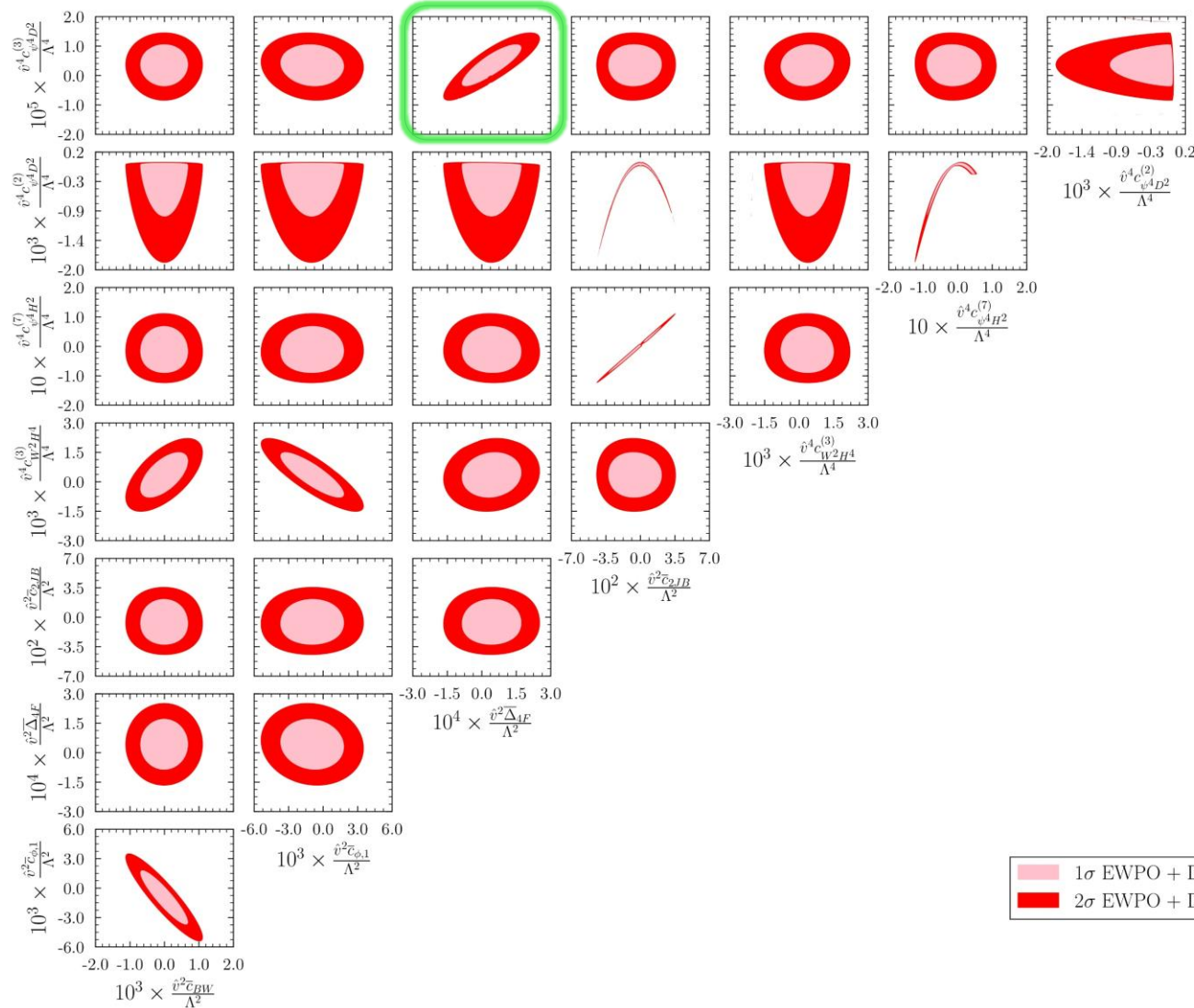


Two-dimensional projections



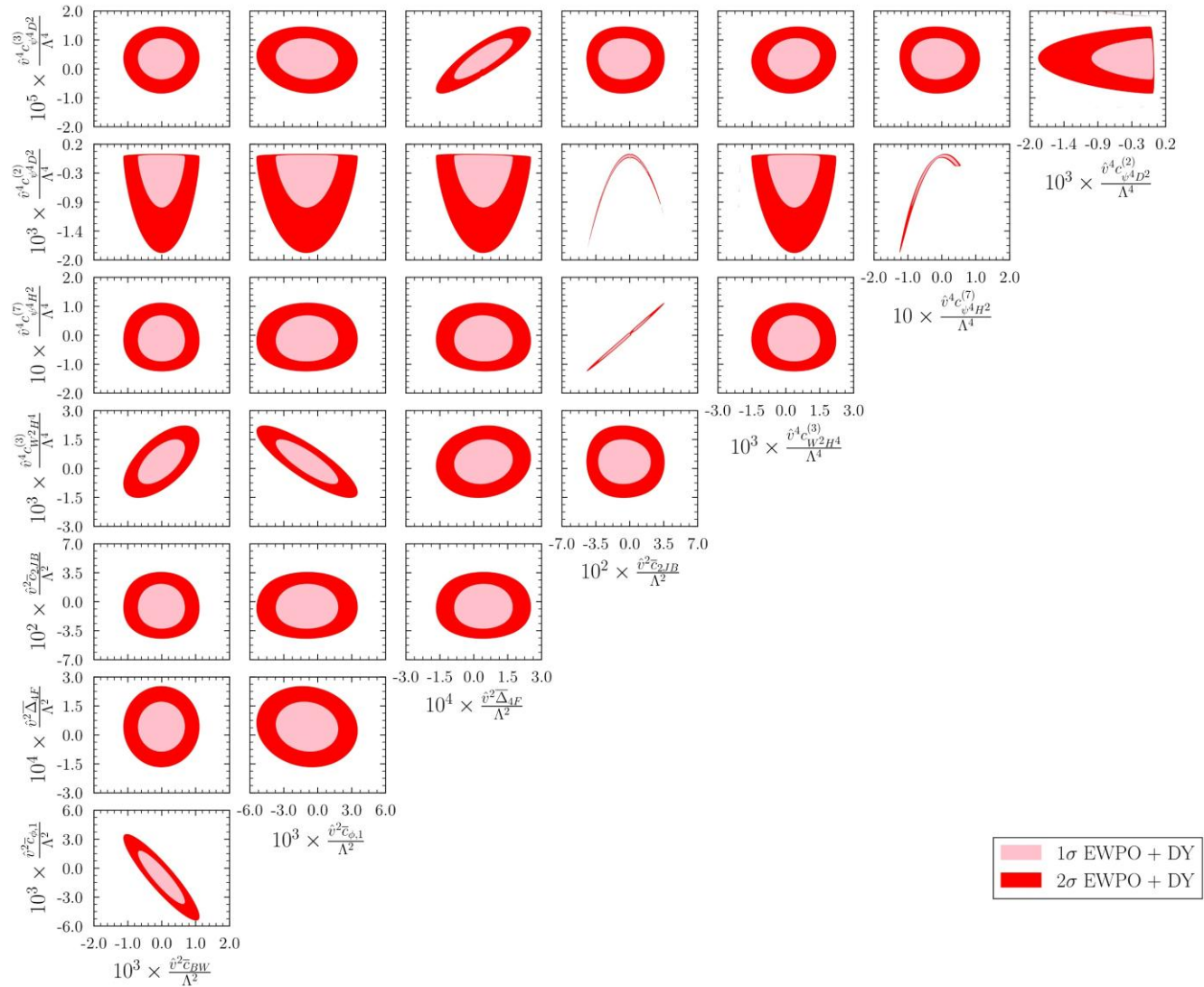
- We find correlations between $\bar{c}_{BW}$ and $\bar{c}_{\Phi,1}$, between $\bar{c}_{\Phi,1}$ and $\bar{c}_{W^2 H^4}^{(3)}$ and between $\bar{c}_{BW}$ and $\bar{c}_{W^2 H^4}^{(3)}$.
- They are related to how they enter in the Electroweak precision observables.
- Along the direction of the correlations, the constraints mainly come from Γ_W which is less precisely determined

Two-dimensional projections

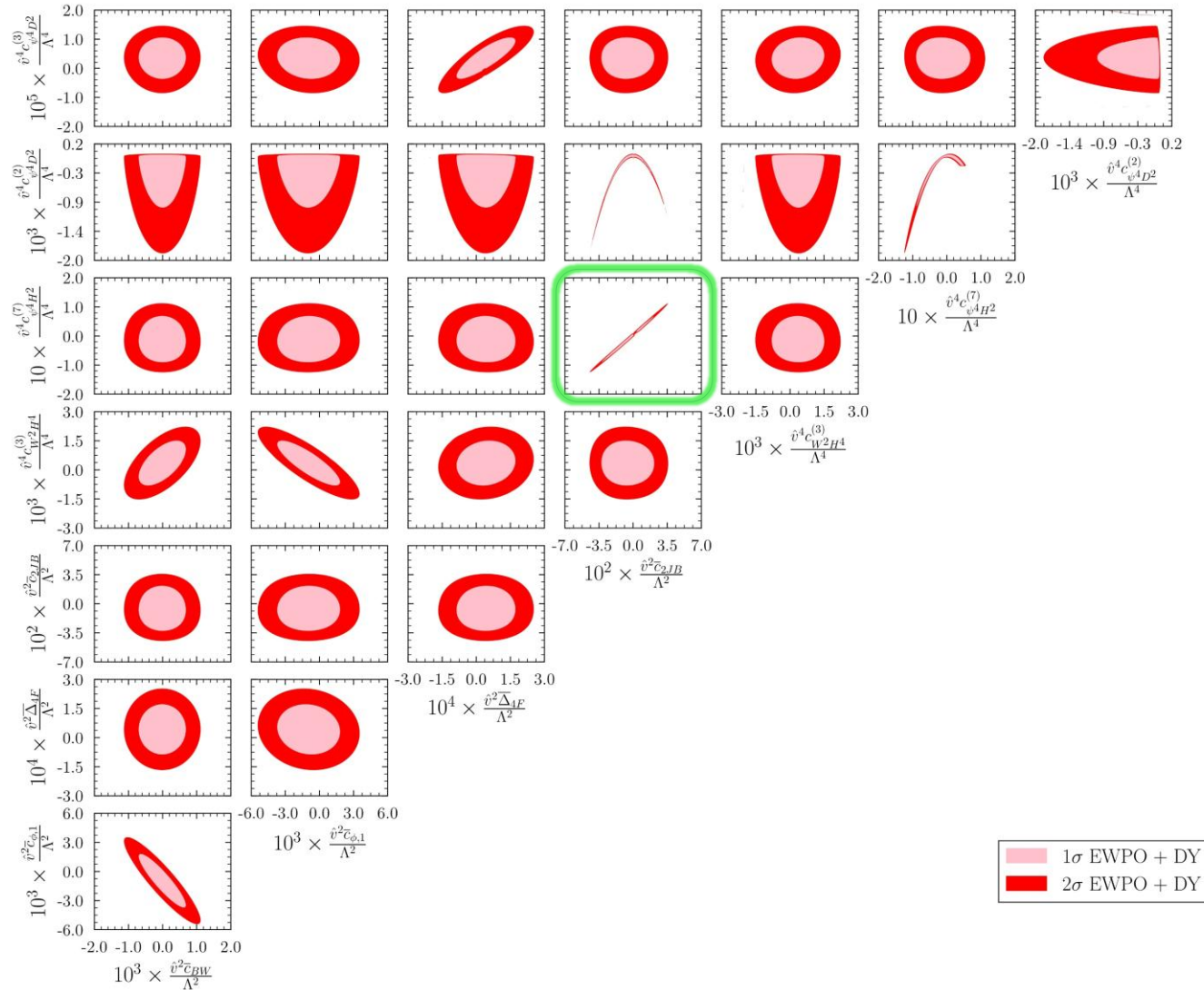


- We find correlations between $\bar{c}_{BW}$ and $\bar{c}_{\Phi,1}$, between $\bar{c}_{\Phi,1}$ and $\bar{c}_{W^2 H^4}^{(3)}$ and between $\bar{c}_{BW}$ and $\bar{c}_{W^2 H^4}^{(3)}$.
- They are related to how they enter in the Electroweak precision observables.
- Along the direction of the correlations, the constraints mainly come from Γ_W which is less precisely determined
- The moderate correlation between $\bar{\Delta}_{4F}(\bar{c}_{2JW})$ and $\bar{c}_{\psi^4 D^2}^{(3)}$ comes from both EWPO and four-fermion contact DY.

Two-dimensional projections

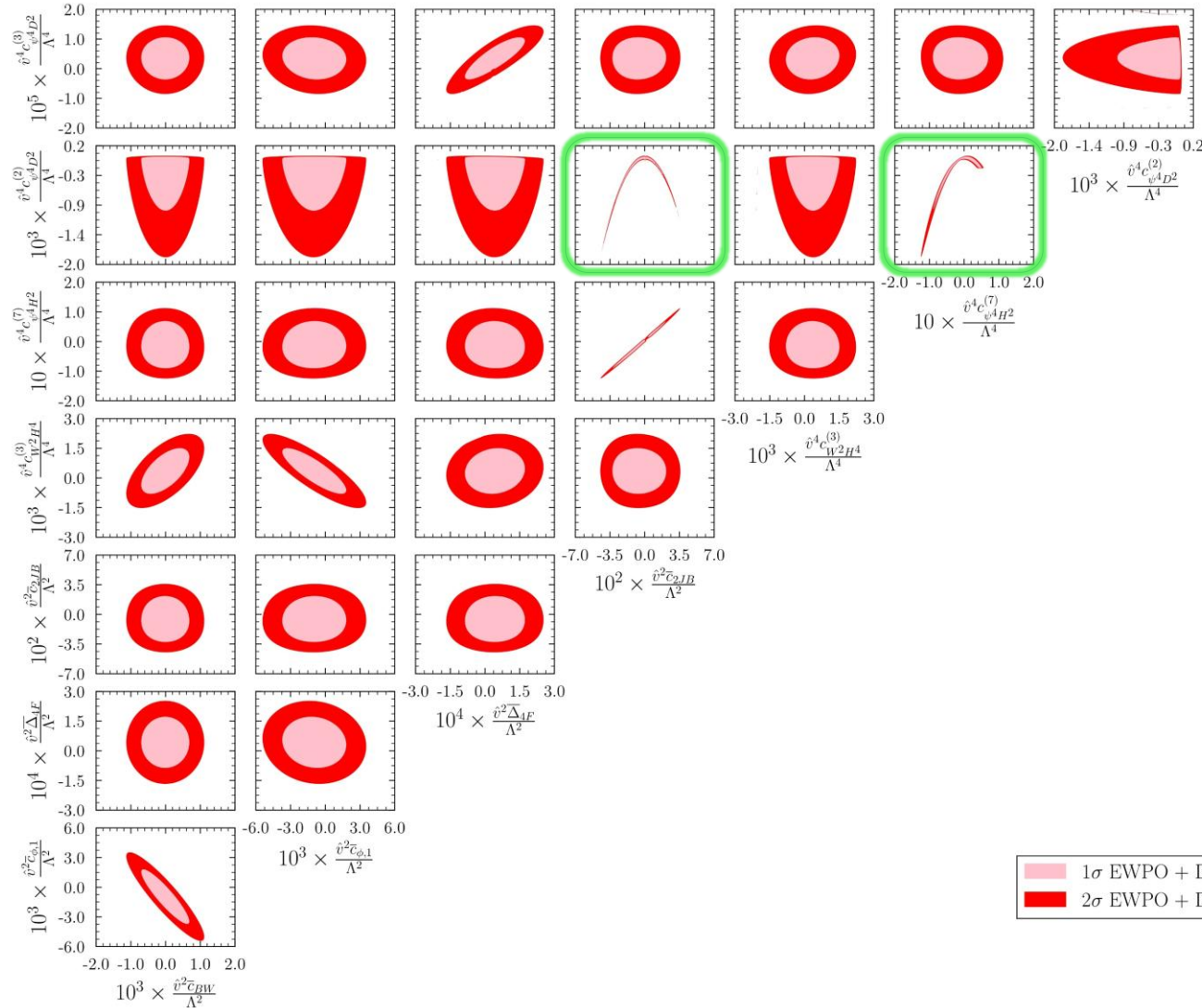


Two-dimensional projections



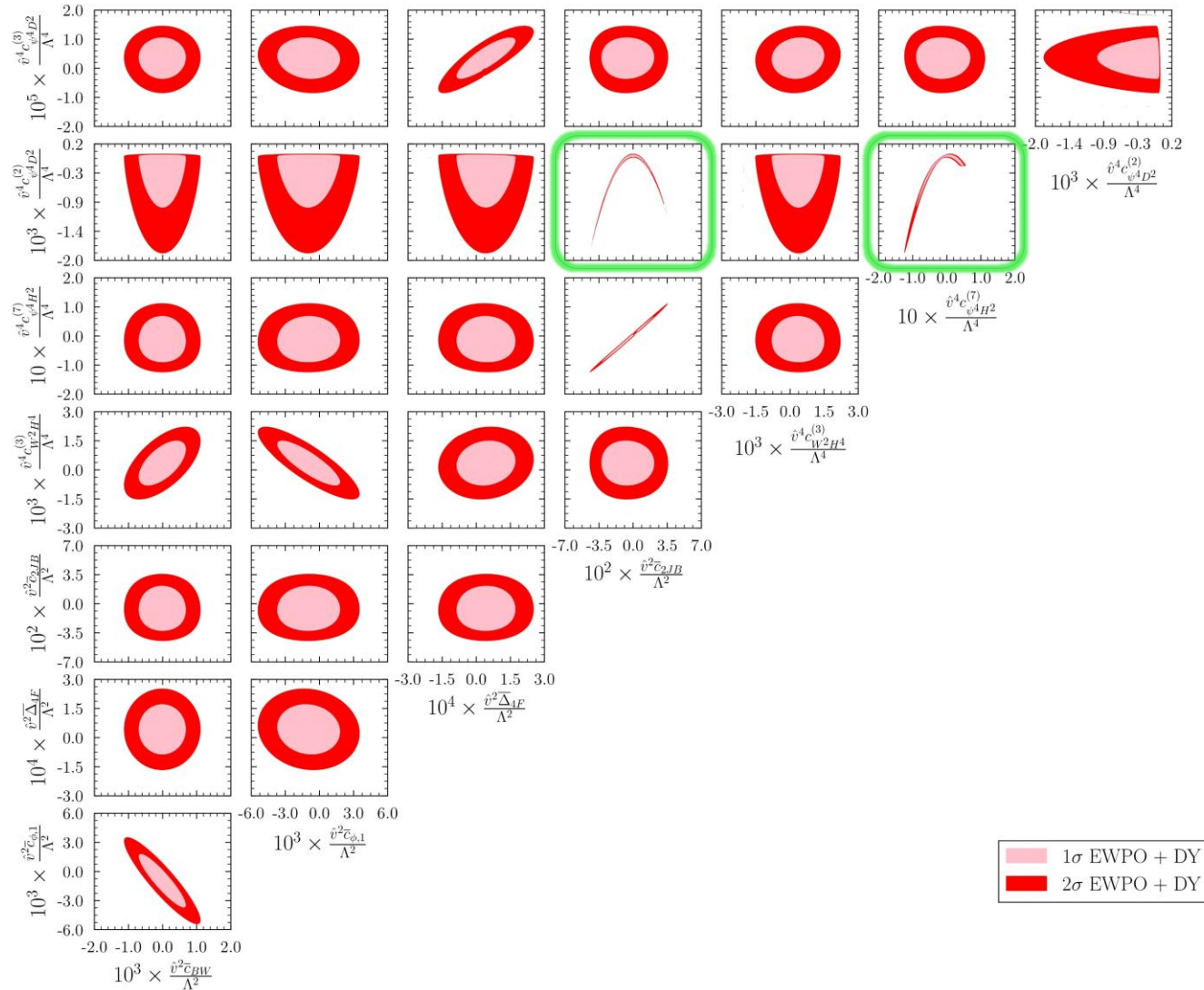
- We also observe a very strong correlation between $\bar{c}_{2JB}$ and $c_{\psi^4 H^2}^{(7)}$.
- This is due to cancelation in the four-fermion contact term $\bar{\mathcal{N}}_{\gamma\gamma}$, which is numerically more relevant.

Two-dimensional projections



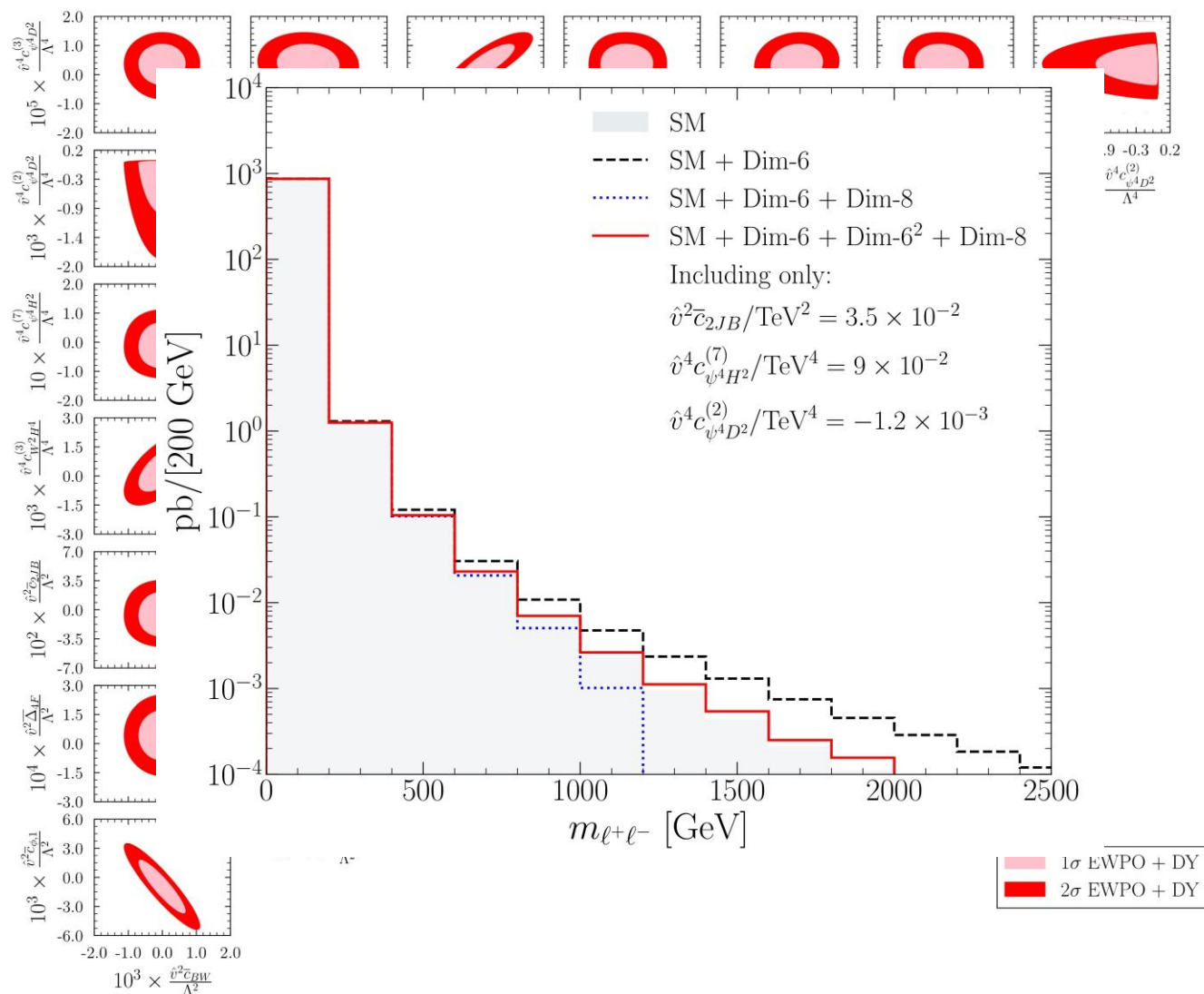
- We also observe a very strong correlation between $\bar{c}_{2JB}$ and $c_{\psi^4 H^2}^{(7)}$.
- This is due to cancelation in the four-fermion contact term $\bar{\mathcal{N}}_{\gamma\gamma}$, which is numerically more relevant.
- We find highly correlated non-elliptical regions for $\bar{c}_{2JB}$ and $c_{\psi^4 D^2}^{(2)}$ and also for $c_{\psi^4 H^2}^{(7)}$ and $c_{\psi^4 D^2}^{(2)}$.
- These cancelations occur for negative values for $c_{\psi^4 D^2}^{(2)}$ when quadratic terms of $\bar{c}_{2JB}$ are included.

Two-dimensional projections



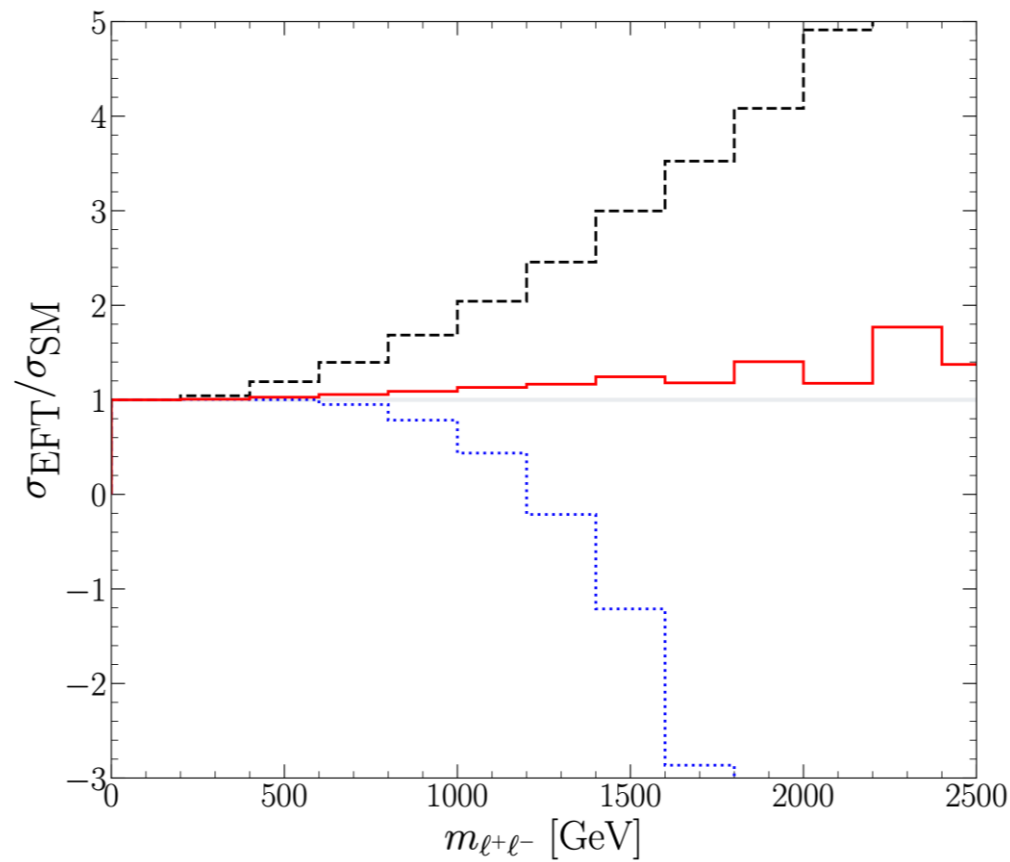
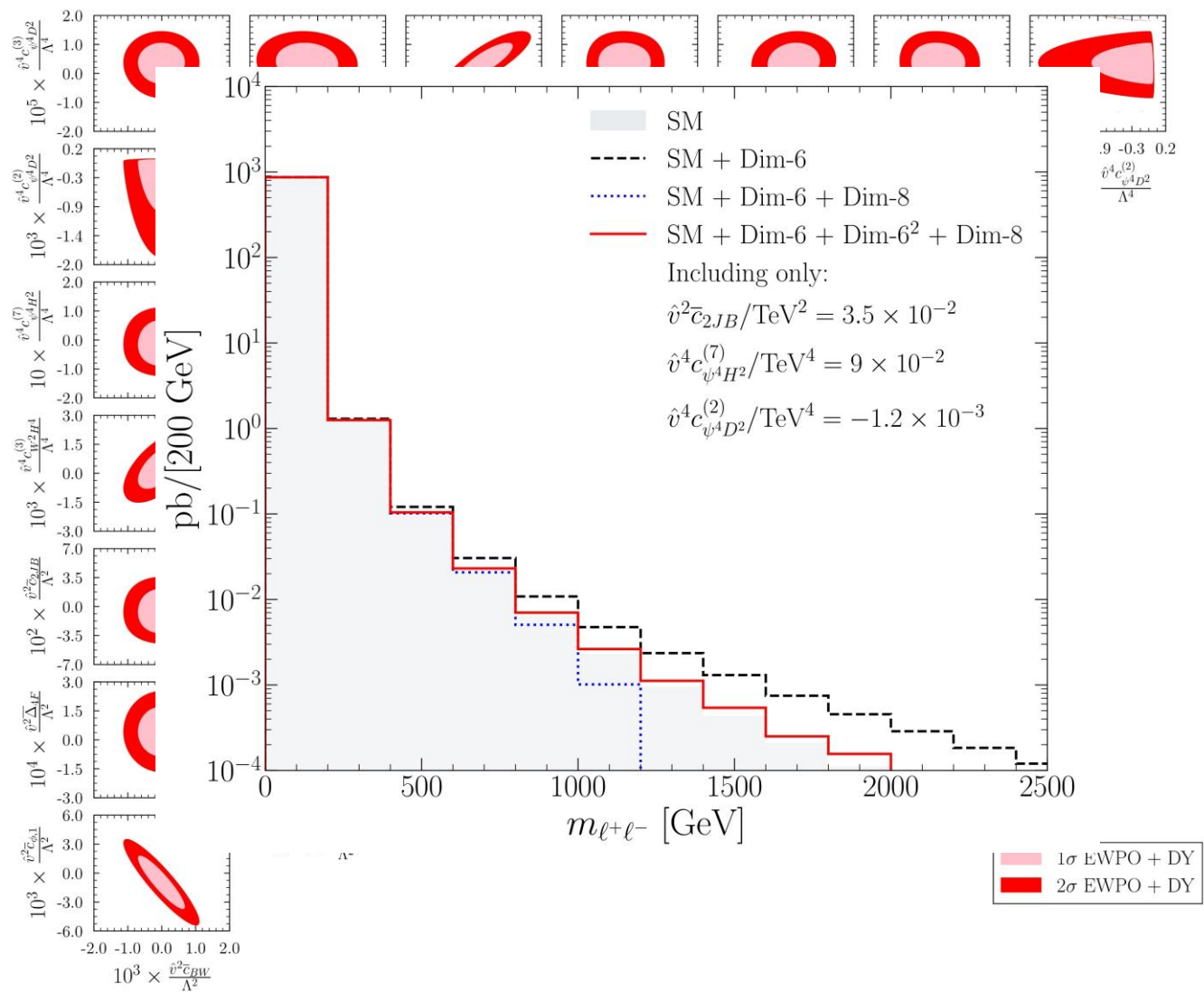
- We also observe a very strong correlation between $\bar{c}_{2JB}$ and $c_{\psi^4 H^2}^{(7)}$.
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Two-dimensional projections

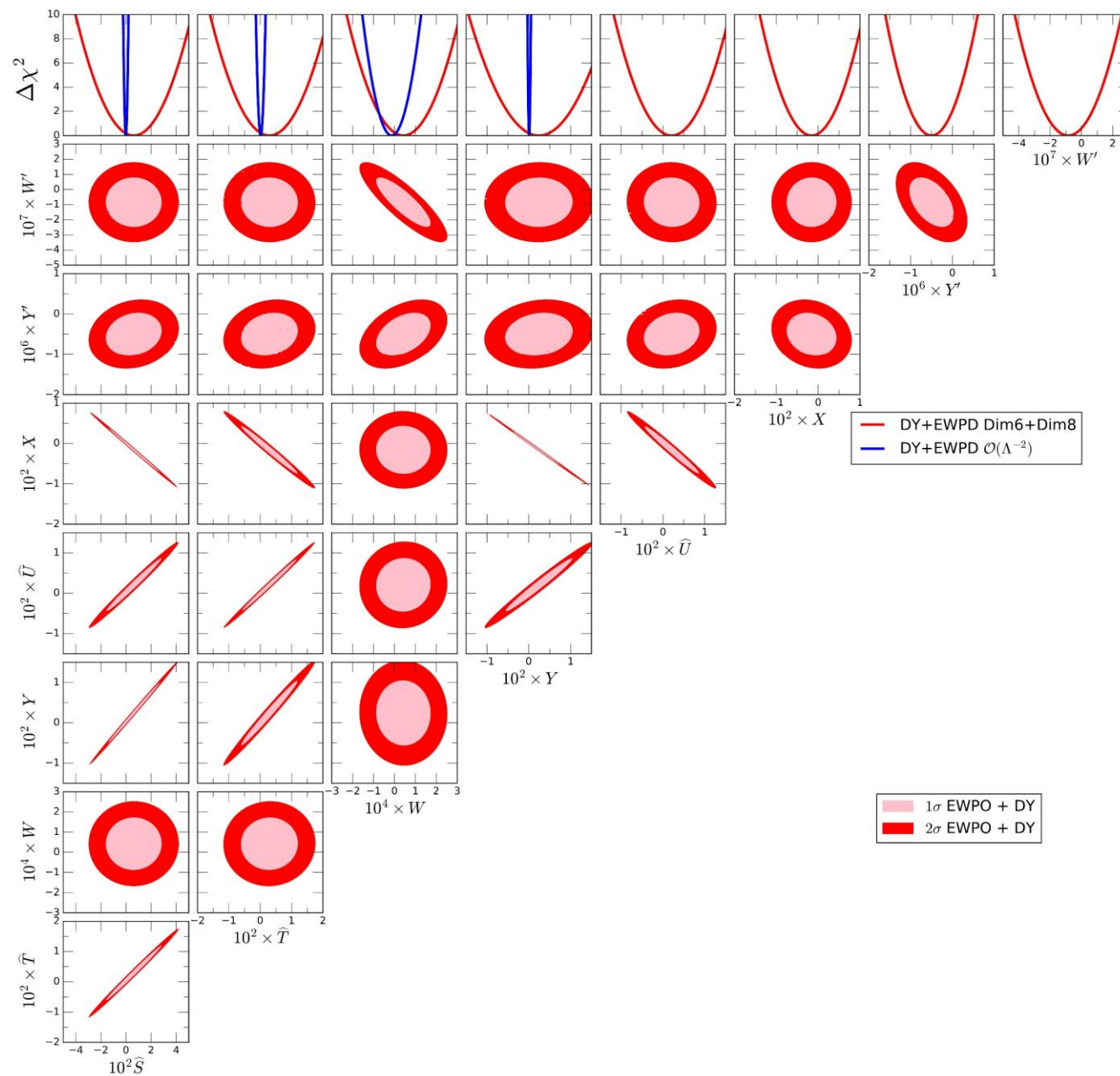


- We also observe a very strong correlation between $\bar{c}_{2JB}$ and $c_{\psi^4 H^2}^{(7)}$.
- This is due to cancelation in the four-fermion contact term $\overline{\mathcal{N}}_{\gamma\gamma}$, which is numerically more relevant.
- We find highly correlated non-elliptical regions for $\bar{c}_{2JB}$ and $c_{\psi^4 D^2}^{(2)}$ and also for $c_{\psi^4 H^2}^{(7)}$ and $c_{\psi^4 D^2}^{(2)}$.
- These cancelations occur for negative values for $c_{\psi^4 D^2}^{(2)}$ when quadratic terms of $\bar{c}_{2JB}$ are included.

Two-dimensional projections



Oblique parameters results



W & Y parameters

Dimension 6

Q_{2JW}	$J_{W\mu}^I J_W^{I\mu}$	c_{2JW}
Q_{2JB}	$J_{B\mu} J_B^\mu$	c_{2JB}

W



Y

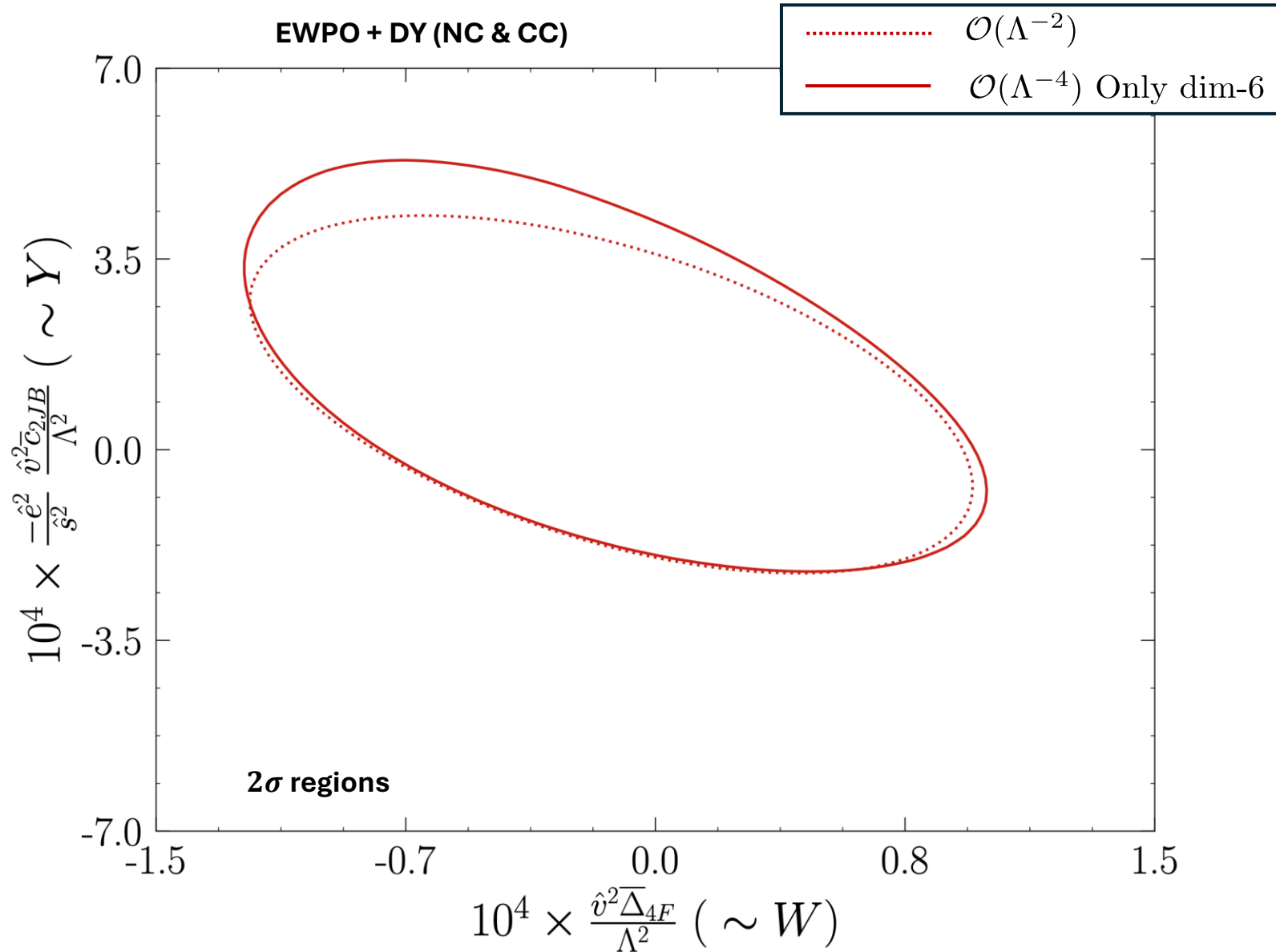
$$J_B^\mu = g' \sum_{f \in \{q, l, u, d, e\}} \sum_a Y_f \bar{f}_a \gamma^\mu f_a ,$$

$$J_W^{I\mu} = \frac{g}{2} \sum_{f \in \{q, l\}} \sum_a \bar{f}_a \gamma^\mu \tau^I f_a$$

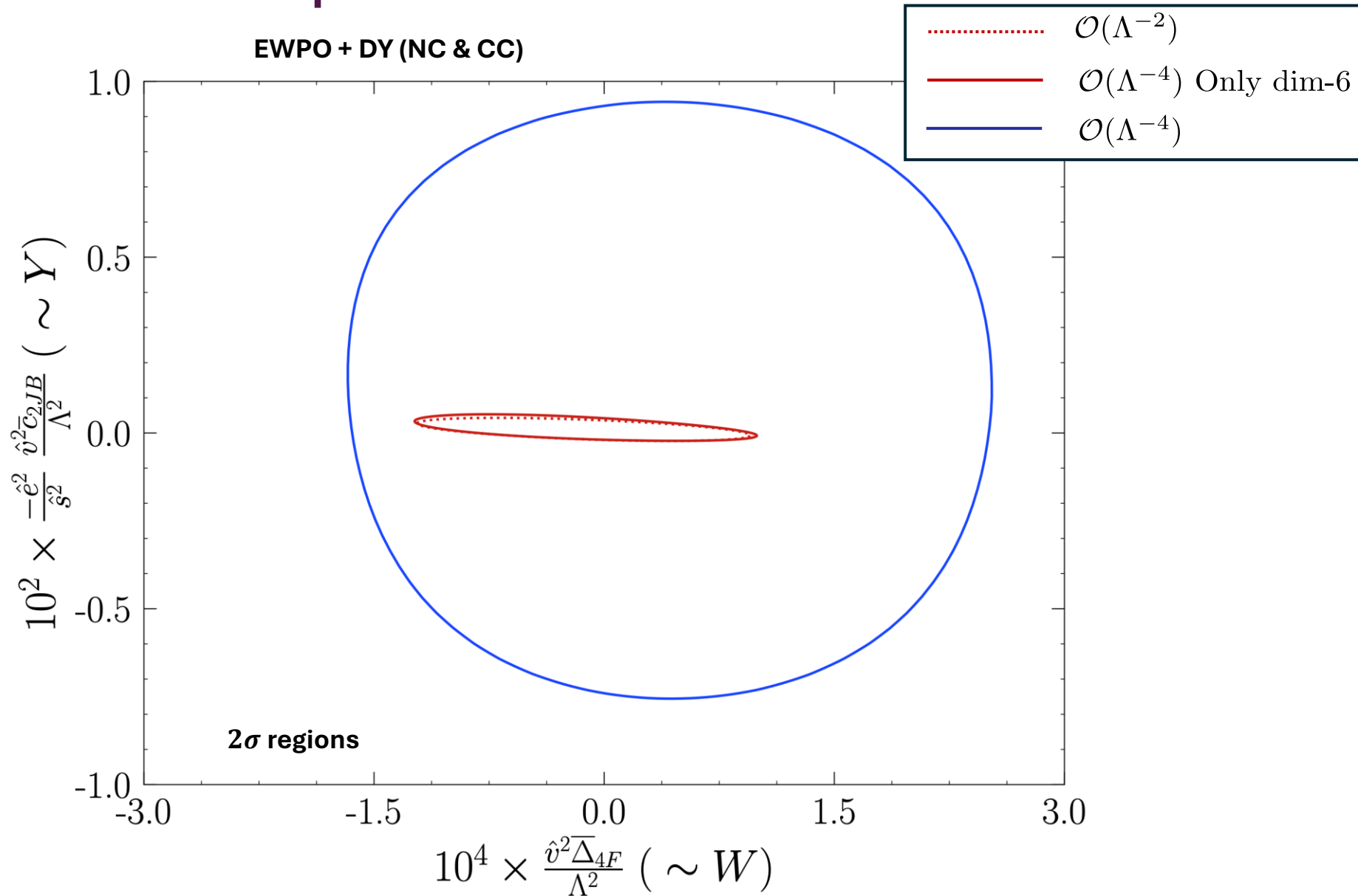
Dimension 8

$Q_{\psi^4 H^2}^{(4)}$	$J_B^\mu J_{B\mu} (H^\dagger H)$	$c_{\psi^4 H^2}^{(4)}$
$Q_{\psi^4 H^2}^{(5)}$	$J_W^{I\mu} J_{W\mu}^I (H^\dagger H)$	$c_{\psi^4 H^2}^{(5)}$

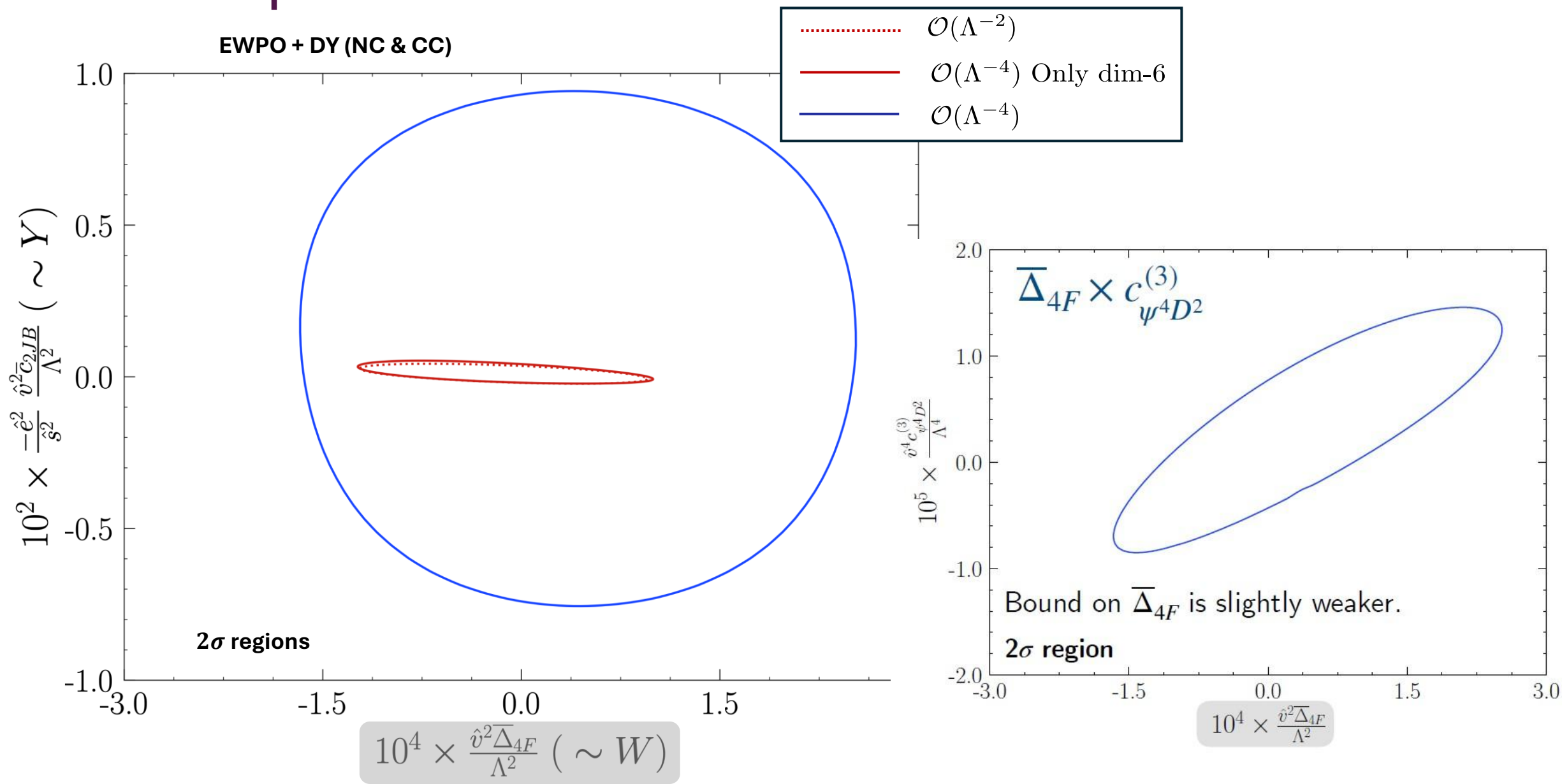
W & Y parameters



W & Y parameters

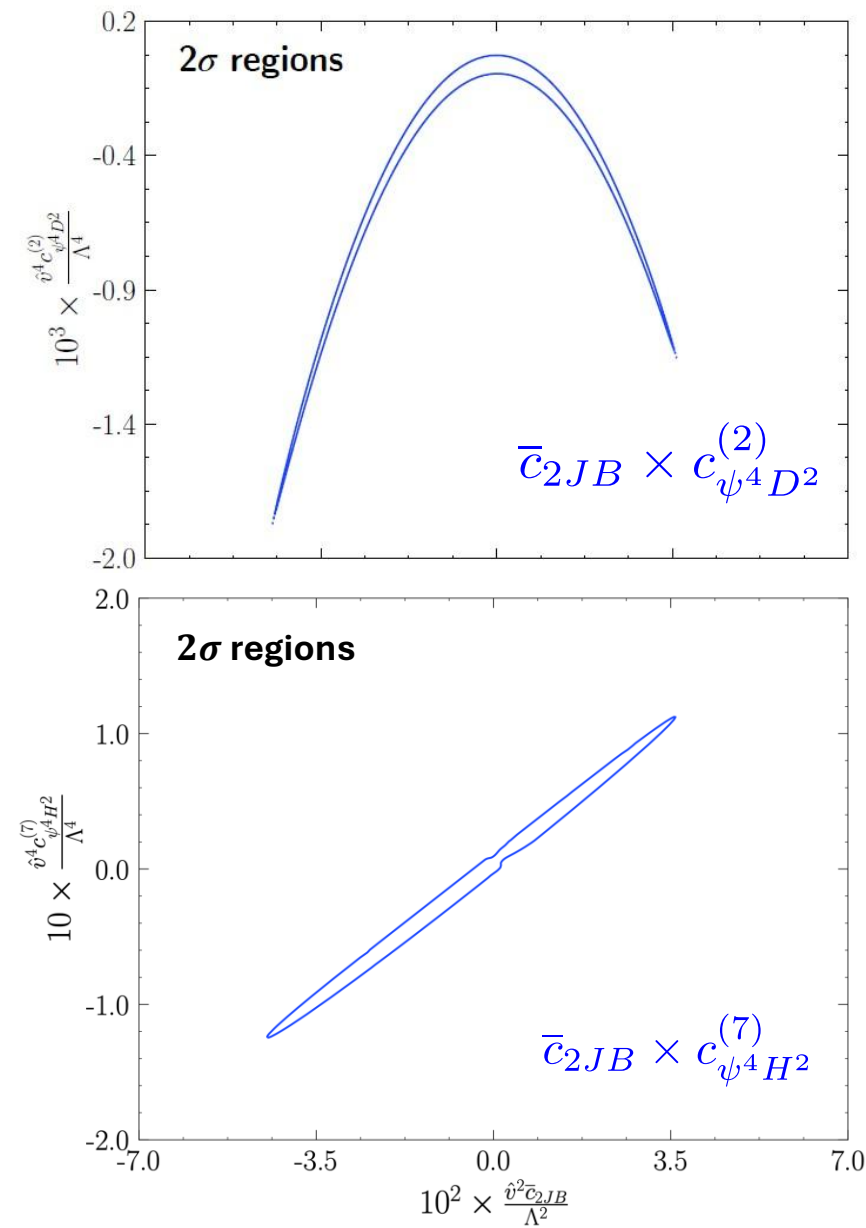
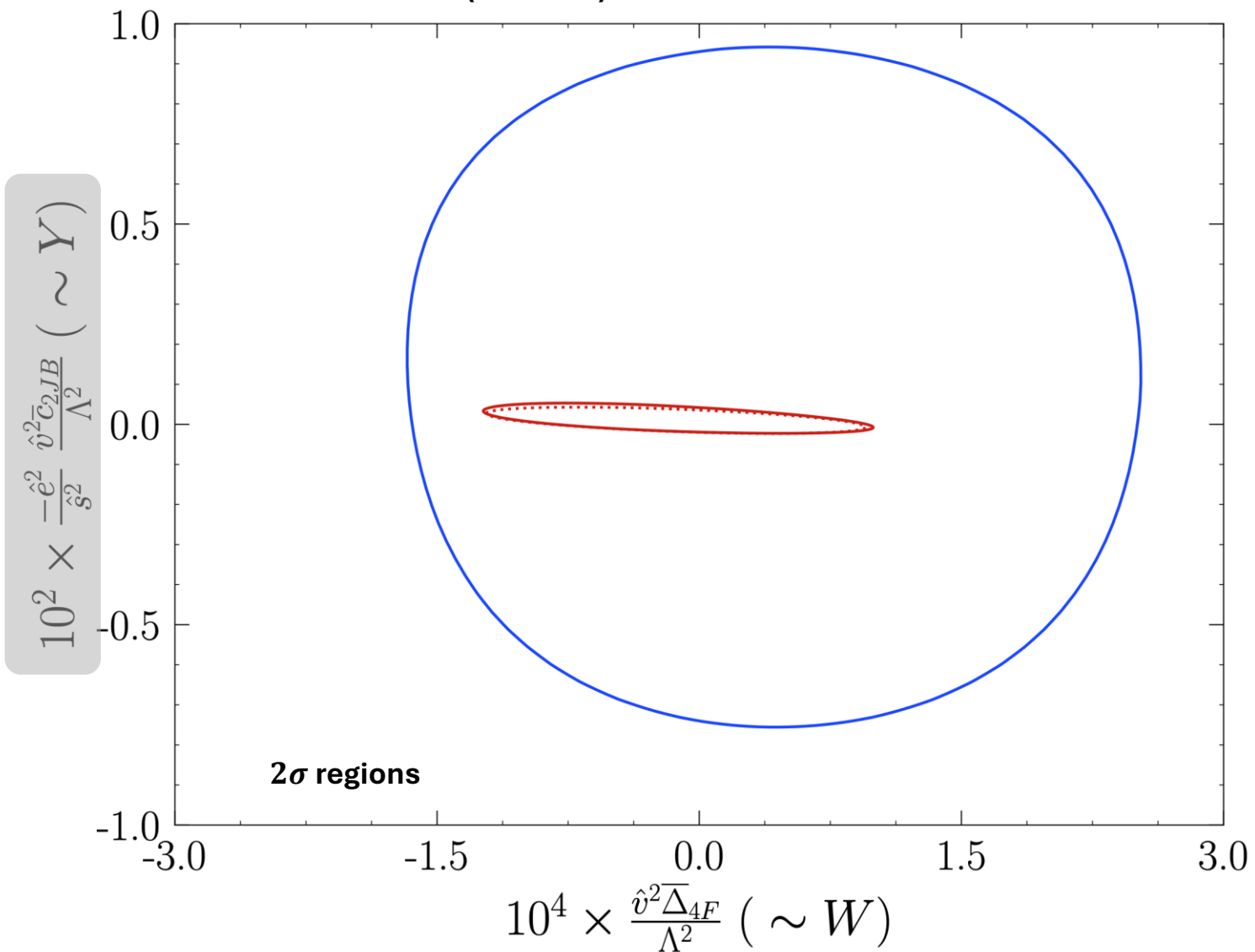


W & Y parameters

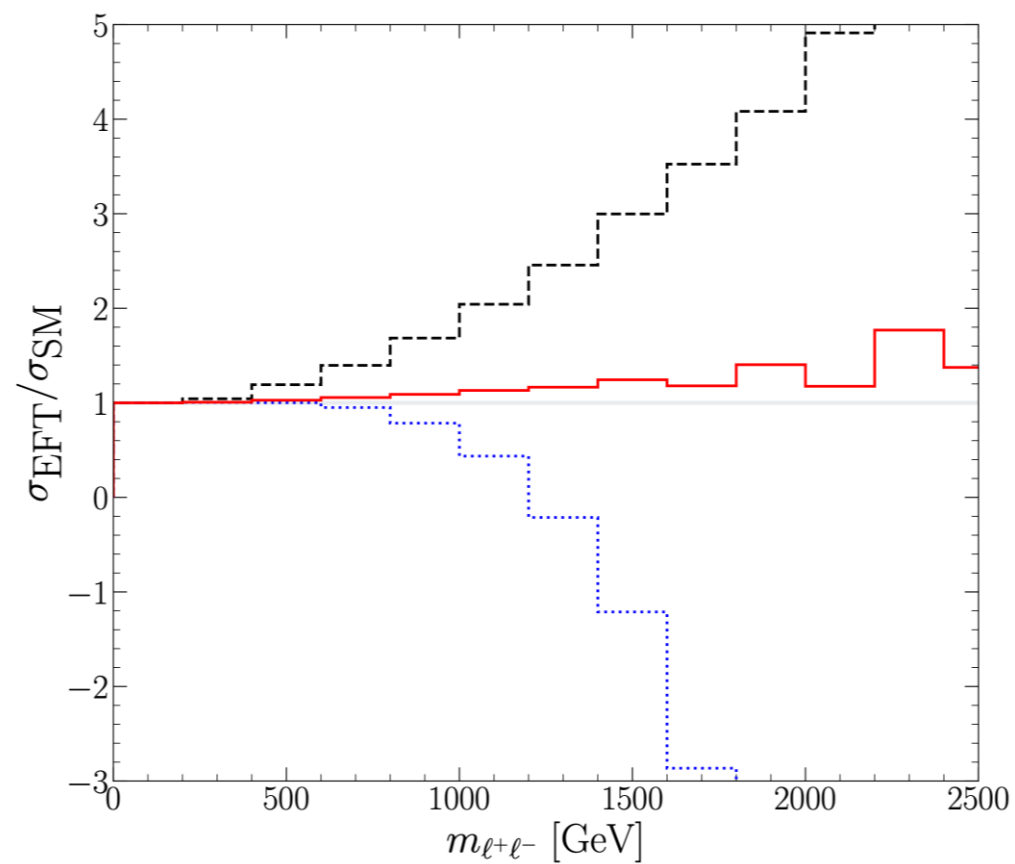
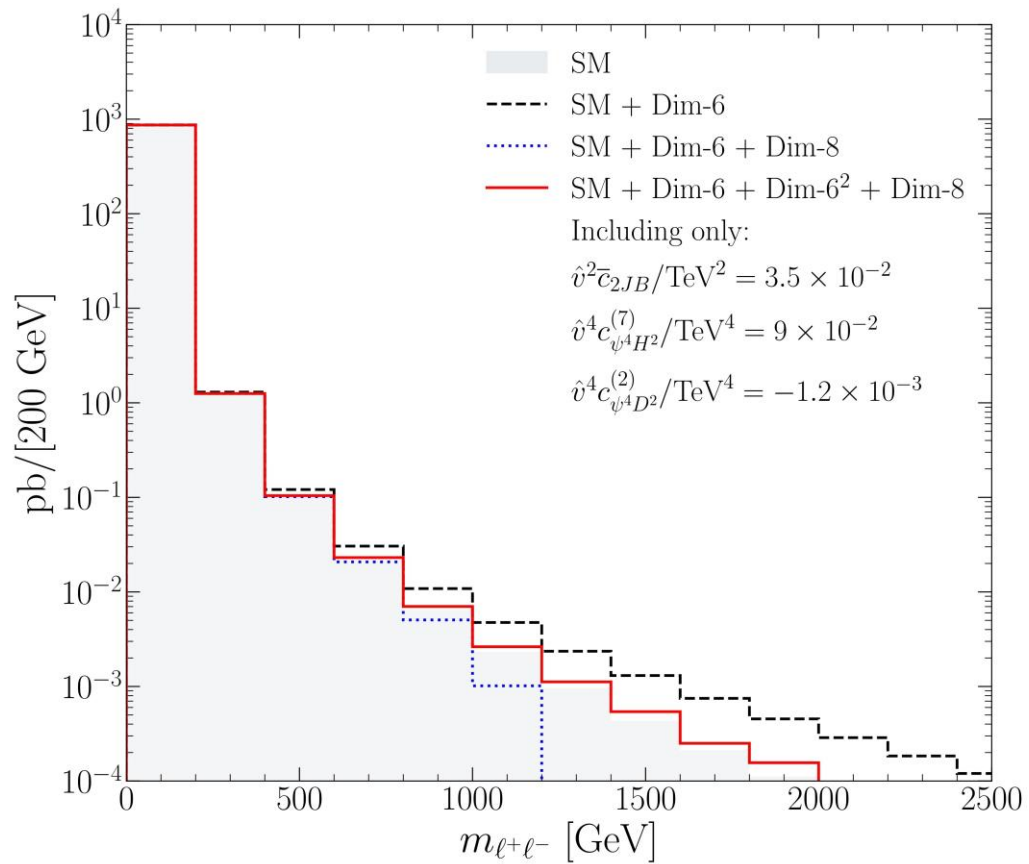


W & Y parameters

EWPO + DY (NC & CC)



EFT truncation



Summary

- We constructed the dimension-eight Universal SMEFT (USMEFT) operator basis which encodes the low-energy effects of Universal extensions of SM.
- We have studied constraints from Drell-Yan production and Electroweak precision in universal theories consistently including effects beyond leading order $\mathcal{O}(\Lambda^{-2})$.
- We also introduced an extension of the parametrization of universal effects in terms of oblique parameters obtained by linearly expanding the self-energies of the electroweak gauge-bosons to $\mathcal{O}(q^6)$.
- Our results show that the present data, that favors no deviation from the SM, can robustly constrain six out of the eight parameters while strong correlations and cancellations are present for the other combinations.

Thank you!

Jay Desai
YITP, Stony Brook University.

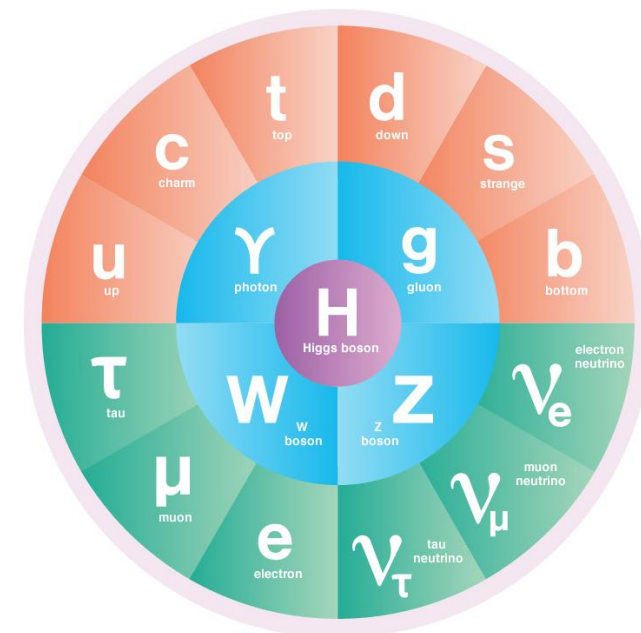


C.N. YANG INSTITUTE
for Theoretical Physics

Backup

Standard Model

$$\begin{aligned}\mathcal{L}_{\text{SM}} = & -\frac{1}{4}G_{\mu\nu}^A G^{A\mu\nu} - \frac{1}{4}W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4}B_{\mu\nu} B^{\mu\nu} \\ & + \sum_{\psi} \bar{\psi} i \not{D} \psi + (D_{\mu} H)^{\dagger} (D^{\mu} H) - \mu^2 H^{\dagger} H - \lambda (H^{\dagger} H)^2 \\ & - (\bar{Q}_L Y_u \tilde{H} u_R + \bar{Q}_L Y_d H d_R + \bar{L}_L Y_e H e_R + \text{h.c.}).\end{aligned}$$



$$SU(3)_C \otimes SU(2)_L \otimes U(1)_Y$$

- Can not explain neutrino masses, matter anti-matter asymmetry, Dark Matter, and Dark Energy.

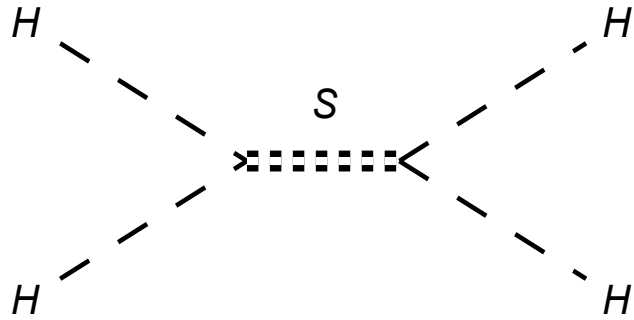
Standard Model Effective Field Theory

- Under the assumption, that the scalar particle discovered in 2012 is part of an electroweak doublet, the $SU(2)_L \otimes U(1)_Y$ symmetry can be linearly realized.

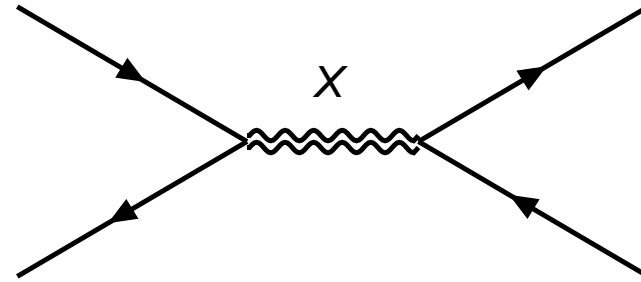
$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_{n>4,j} \frac{f_{n,j}}{\Lambda^{n-4}} \mathcal{O}_{n,j} ,$$

- In SMEFT, the number of possible dimension eight operators are 44,807.
- Identifying unique hypotheses that capture a sizable class of Beyond Standard Model theories while reducing the number of operators becomes important.
- One such hypothesis is universality.

Simple UV Completions for universal theories

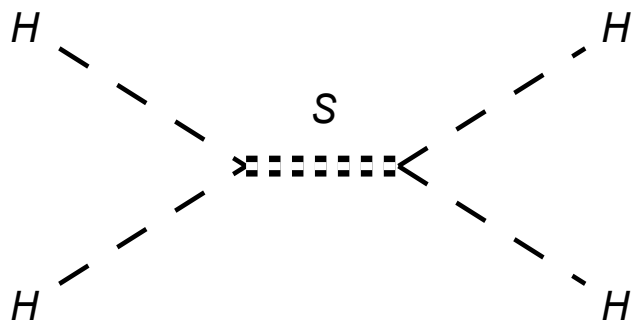


- S is a heavy scalar. It can be a real scalar singlet or a triplet under $SU(2)_L$.
- The scalar field does not couple to the Standard Model fermions.



- X is a heavy vector boson. It can be charged under a new $U(1)'$.
- The couplings to Standard Model fermions are via SM currents.

Matching models: a real scalar singlet



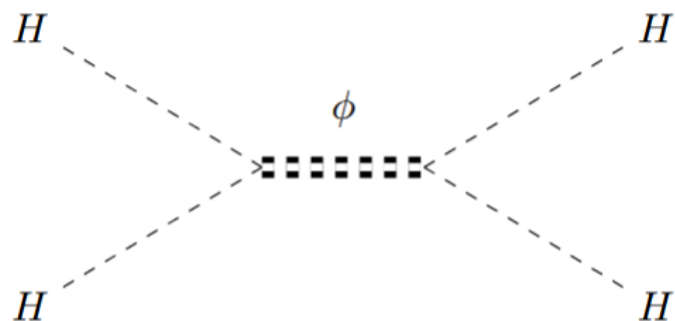
Integrate out heavy S and match on to the *rotated* basis.

$$\Delta\mathcal{L} = \frac{1}{2}(\partial_\mu S)^2 - \frac{1}{2}M^2 S^2 - A|H|^2 S - \frac{1}{2}k|H|^2 S^2 - \frac{1}{3!}\mu S^3 - \frac{1}{4!}\tilde{\lambda}_S S^4$$

S is a real scalar singlet.

$$\begin{aligned} Q_{\psi^2 H^5}^{(1)} &: (H^\dagger H)^2 (J_H H) \\ Q_{\psi^2 H^3 D^2}^{(1)} &: (D_\mu H^\dagger D^\mu H) (J_H H) \\ Q_{\psi^4 H^2}^{(1)} &: J_H J_H^\dagger (H^\dagger H) \\ Q_{\psi^4 H^2}^{(2)} &: J_H^j J_H^k (H_j H_k) \\ Q_{\psi^4 H^2}^{(3)} &: H^{\dagger k} J_H^j J_{Hk}^\dagger H_j \end{aligned}$$

Matching models: Scalar triplet



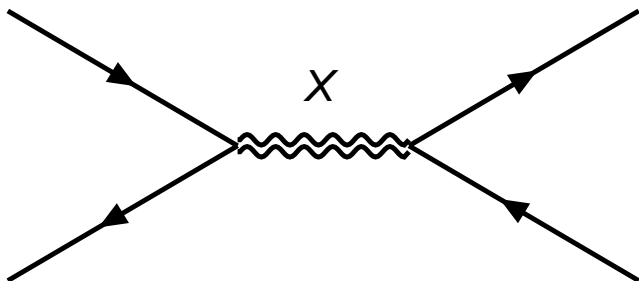
Integrate out heavy ϕ and match on to the *rotated* basis.

$$\Delta\mathcal{L} = \frac{1}{2}(D_\mu\phi^a)(D^\mu\phi^a) - \frac{1}{2}M^2(\phi^a)^2 + \kappa H^\dagger\sigma^a H\phi^a - \lambda_{\phi H}(\phi^a)^2|H|^2 - \lambda_\phi|\phi^a|^4$$

ϕ^a is a triplet under $SU(2)_L$.

$$\begin{aligned} Q_{\psi^2 H^5}^{(1)} &: (H^\dagger H)^2 (J_H H) \\ Q_{\psi^2 H^3 D^2}^{(2)} &: (D_\mu H^\dagger \tau^I D^\mu H) (J_H \tau^I H) \\ Q_{\psi^4 H^2}^{(1)} &: J_H J_H^\dagger (H^\dagger H) \\ Q_{\psi^4 H^2}^{(2)} &: J_H^j J_H^k (H_j H_k) \\ Q_{\psi^4 H^2}^{(3)} &: H^{\dagger k} J_H^j J_{Hk}^\dagger H_j \end{aligned}$$

Matching models: U(1)' gauge boson



Integrate out heavy X and match on to the *rotated* basis.

$$\Delta\mathcal{L} = -\frac{1}{4}X_{\mu\nu}X^{\mu\nu} + \frac{1}{2}M^2X_\mu X^\mu - \frac{k}{2}B_{\mu\nu}X^{\mu\nu}$$

$X_{\mu\nu}$ is charged under $U(1)'$.

$$\begin{aligned} Q_{\psi^2 H^5}^{(1)} &: (H^\dagger H)^2 (J_H H) \\ Q_{\psi^2 H^3 D^2}^{(1)} &: (D_\mu H^\dagger D^\mu H) (J_H H) \\ Q_{\psi^4 H^2}^{(1)} &: J_H J_H^\dagger (H^\dagger H) \\ Q_{\psi^4 H^2}^{(2)} &: J_H^j J_H^k (H_j H_k) \\ Q_{\psi^4 H^2}^{(3)} &: H^{\dagger k} J_H^j J_{Hk}^\dagger H_j \\ Q_{\psi^4 D^2}^{(4)} &: D^\alpha J_G^{A\mu} D_\alpha J_{G\mu}^A \\ Q_{\psi^2 H^4 D}^{(2)} &: i J_W^{I\mu} \left[(H^\dagger \overleftrightarrow{D}_\mu^I H) (H^\dagger H) + (H^\dagger \overleftrightarrow{D}_\mu H) (H^\dagger \tau^I H) \right] \end{aligned}$$

Phenomenology with the *rotated* basis

- We have the input parameters

$$\{\hat{\alpha}_{\text{em}} , \hat{G}_F , \hat{M}_Z\}$$

$$\begin{aligned} \hat{e} &= \sqrt{4\pi\hat{\alpha}_{\text{em}}} , & \hat{c}^2 \hat{s}^2 &= \frac{\pi\hat{\alpha}_{\text{em}}}{\sqrt{2} \hat{G}_F \hat{M}_Z^2} , \\ \hat{v}^2 &= \frac{1}{\sqrt{2} \hat{G}_F} , \end{aligned}$$

- Higher dimension operators will induce renormalization of the Standard Model couplings g', g, v .

EWPO and TGC analysis

- In our first work⁺, we studied the impact of Universal dimension eight operators on Electroweak Precision Observables and Triple Gauge Couplings.

⁺T. Corbett, J. **Desai**, O. J. P. Éboli, M. C. Gonzalez-Garcia, M. Martines, and P. Reimitz
arXiv:2304.03305[hep-ph]

- The relevant bosonic operators at both dimension-six* and dimension-eight**

Dimension 6

$$\begin{aligned} Q_{\Phi,1} &: (D_\mu H^\dagger H)(H^\dagger D^\mu H) \\ Q_{BB} &: H^\dagger B_{\mu\nu} B^{\mu\nu} H \\ Q_{WW} &: H^\dagger W_{\mu\nu}^I W^{I,\mu\nu} H \\ Q_{BW} &: H^\dagger B_{\mu\nu} \tau^I W^{I,\mu\nu} H \end{aligned}$$

Dimension 8

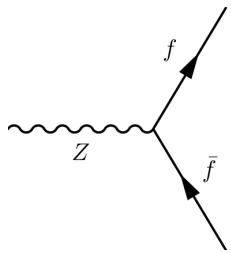
$$\begin{aligned} Q_{W^2 H^4}^{(1)} &: (H^\dagger H)^2 W_{\mu\nu}^I W^{I\mu\nu} & Q_{H^6}^{(1)} &: (H^\dagger H)^2 (D_\mu H^\dagger D^\mu H) \\ Q_{B^2 H^4}^{(1)} &: (H^\dagger H)^2 B_{\mu\nu} B^{\mu\nu} & Q_{H^6}^{(2)} &: (H^\dagger H)(H^\dagger \tau^I H)(D_\mu H^\dagger \tau^I D^\mu H) \\ Q_{WBH^4}^{(1)} &: (H^\dagger H)(H^\dagger \tau^I H) W_{\mu\nu}^I B^{\mu\nu} \\ Q_{W^2 H^4}^{(3)} &: (H^\dagger \tau^I H)(H^\dagger \tau^J H) W_{\mu\nu}^I W^{J\mu\nu} \end{aligned}$$

*J.D. Wells and Z. Zhang arxiv:1510.08462 [hep-ph]

**C.W. Murphy arXiv:2005.00059 [hep-ph]

Electro-Weak Precision Observables

- EWPO, where we include 12 Z-pole observables, and 2 W-pole observables:



$\Gamma_Z, \sigma_h^0, \mathcal{A}_\ell(\tau^{\text{pol}}), R_\ell^0, \mathcal{A}_\ell(\text{SLD}), A_{\text{FB}}^{0,l}, R_c^0, R_b^0, \mathcal{A}_c, \mathcal{A}_b, A_{\text{FB}}^{0,c}, \text{ and } A_{\text{FB}}^{0,b}$

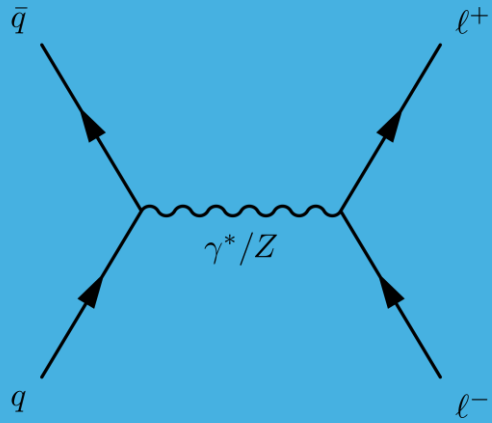
M_W and Γ_W

- We adopt as input parameters

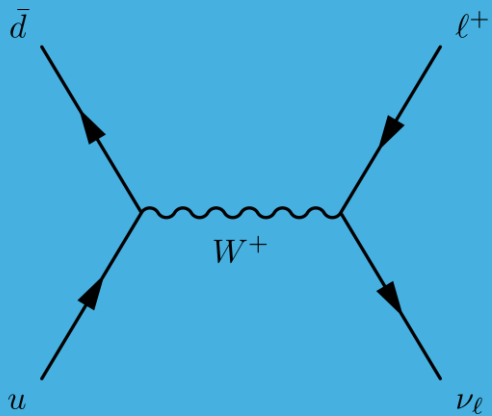
$$\{\hat{\alpha}_{\text{em}}, \hat{G}_F, \hat{M}_Z\}$$

$$\begin{aligned} \hat{e} &= \sqrt{4\pi\hat{\alpha}_{\text{em}}} , & \hat{c}^2\hat{s}^2 &= \frac{\pi\hat{\alpha}_{\text{em}}}{\sqrt{2}\hat{G}_F\hat{M}_Z^2} , \\ \hat{v}^2 &= \frac{1}{\sqrt{2}\hat{G}_F} , \end{aligned}$$

Drell Yan

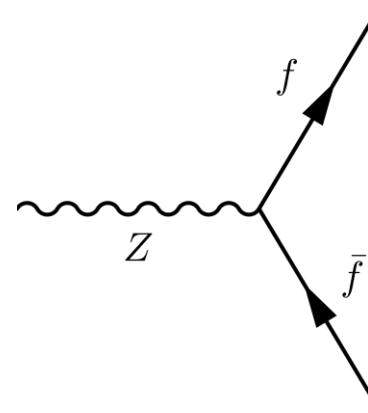


Neutral-Current

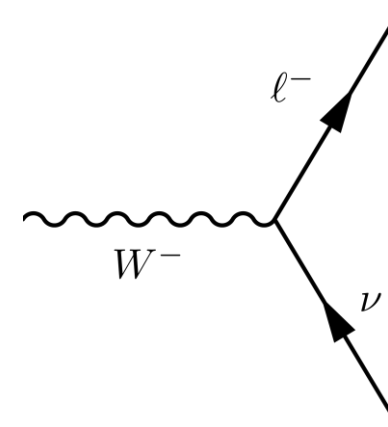


Charged-Current

Electroweak Precision



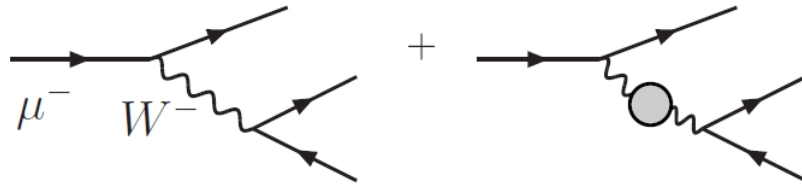
Z-pole
observables



W-pole
observables

Fermionic operators: Finite renormalization of G_F

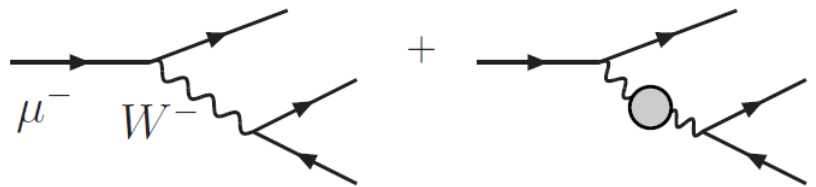
- The muon decay observable $\hat{G}_F$ is computed from the lifetime of the muon,



$$\left[2\langle H^\dagger H \rangle - \frac{1}{\sqrt{2}\hat{G}_F} \right]_{\text{fermionic}} \equiv \frac{\hat{v}^4}{\Lambda^2} \Delta_{4F}^{(6)} + \frac{\hat{v}^6}{\Lambda^4} \Delta_{4F}^{(8)} ,$$

Fermionic operators: Finite renormalization of G_F

- The muon decay observable $\hat{G}_F$ is computed from the lifetime of the muon,

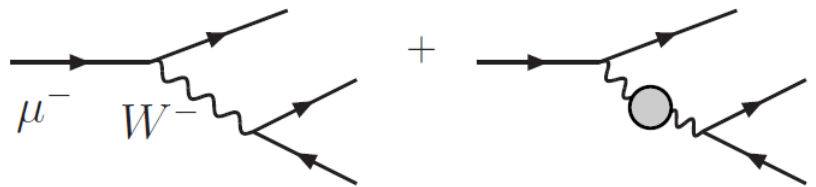


$$\left[2\langle H^\dagger H \rangle - \frac{1}{\sqrt{2}\hat{G}_F} \right]_{\text{fermionic}} \equiv \frac{\hat{v}^4}{\Lambda^2} \Delta_{4F}^{(6)} + \frac{\hat{v}^6}{\Lambda^4} \Delta_{4F}^{(8)} ,$$

$$\Delta_{4F}^{(6)} \sim \mathcal{O}_{2JW} = \sum_{f \in \{q,l\}} \sum_{f' \in \{q,l\}} \left(\bar{f} \gamma^\mu \frac{\tau^I}{2} f \right) \left(\bar{f}' \gamma_\mu \frac{\tau^I}{2} f' \right) .$$

Fermionic operators: Finite renormalization of G_F

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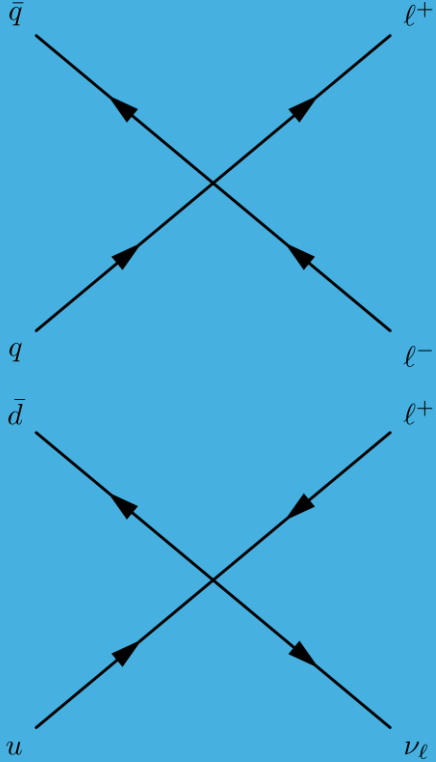


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- We encapsulate all the relevant universal fermionic operators at dimension 8 into a new Wilson coefficient $\Delta_{4F}^{(8)}$.

Four-fermion contact amplitudes



$$\mathcal{M}_{\text{Cont}}^{\text{DY,NC}} = -\hat{e}^2 \frac{1}{\widehat{M}_Z^2} \left[\frac{1}{\hat{s}^2 \hat{c}^2} (j_Z^f)^\mu (j_Z^{f'})_\mu \left(\overline{\mathcal{N}}_{ZZ} + \mathcal{N}_{ZZ}^\square + \frac{p^2}{\widehat{M}_Z^2} \mathcal{N}'_{ZZ} \right) + \frac{1}{\hat{s} \hat{c}} \left((j_Z^f)^\mu (j_Q^{f'})_\mu + (j_Q^f)^\mu (j_Z^{f'})_\mu \right) \left(\overline{\mathcal{N}}_{\gamma Z} + \mathcal{N}_{\gamma Z}^\square + \frac{p^2}{\widehat{M}_Z^2} \mathcal{N}'_{\gamma Z} \right) + (j_Q^f)^\mu (j_Q^{f'})_\mu \left(\overline{\mathcal{N}}_{\gamma\gamma} + \mathcal{N}_{\gamma\gamma}^\square + \frac{p^2}{\widehat{M}_Z^2} \mathcal{N}'_{\gamma\gamma} \right) \right],$$

$$\mathcal{M}_{\text{Cont}}^{\text{DY,CC}} = -\frac{\hat{e}^2}{2\hat{s}^2} \frac{1}{\widehat{M}_W^2} (j_W^f)^\mu (j_W^{f'})_\mu \left(\overline{\mathcal{N}}_{WW} + \mathcal{N}_{WW}^\square + \frac{p^2}{\widehat{M}_W^2} \mathcal{N}'_{WW} \right),$$

- We define the $\overline{\mathcal{N}}$ pieces containing corrections which are dimension-six at leading order with additional irresolvable contributions, $\mathcal{N}^\square$ contains quadratic terms, and $\mathcal{N}'$ contains the coefficients of momentum-dependent four-fermion couplings.

$$\begin{aligned} (j_Z^f)^\mu &= \hat{g}_L^f \bar{f}_L \gamma^\mu f_L + \hat{g}_R^f \bar{f} \gamma^\mu f, \\ (j_Q^f)^\mu &= Q^f \bar{f} \gamma^\mu f, \\ (j_W^f)^\mu &= f_{uL} \gamma^\mu f_{dL}. \end{aligned}$$

$$\begin{aligned}
\overline{\mathcal{N}}_{\gamma\gamma} &= -\frac{\hat{e}^2}{2\hat{s}^2\hat{c}^2} \left(\hat{s}^2\bar{c}_{2JW} + \hat{c}^2\bar{c}_{2JB} \right) \frac{\hat{v}^2}{\Lambda^2} + \frac{\hat{e}^2}{4\hat{s}\hat{c}} c_{\psi^4 H^2}^{(7)} \frac{\hat{v}^4}{2\Lambda^4} , \\
\overline{\mathcal{N}}_{ZZ} &= -\frac{\hat{e}^2}{2\hat{s}^2\hat{c}^2} \left(\hat{c}^2\bar{c}_{2JW} + \hat{s}^2\bar{c}_{2JB} \right) \frac{\hat{v}^2}{\Lambda^2} - \frac{\hat{e}^2}{4\hat{s}\hat{c}} c_{\psi^4 H^2}^{(7)} \frac{\hat{v}^4}{\Lambda^4} , \\
\overline{\mathcal{N}}_{\gamma Z} &= -\frac{\hat{e}^2}{2\hat{s}\hat{c}} \left(\bar{c}_{2JW} - \bar{c}_{2JB} \right) \frac{\hat{v}^2}{\Lambda^2} + \hat{c}_2 \frac{\hat{e}^2}{8\hat{s}^2\hat{c}^2} c_{\psi^4 H^2}^{(7)} \frac{\hat{v}^4}{\Lambda^4} , \\
\overline{\mathcal{N}}_{WW} &= -\frac{\hat{e}^2}{2\hat{s}^2} \bar{c}_{2JW} \frac{\hat{v}^2}{\Lambda^2} .
\end{aligned}$$

Analysis Framework

- We perform a binned log likelihood analysis to study the constraints on the combinations of the USMEFT Wilson coefficients in the *rotated basis* imposed by neutral- and charged-current Drell Yan in combination with EWPO.

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Data can not break the degeneracy.

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$$\Delta g_1^\square = \frac{1}{32} \frac{1}{\hat{c}_2} \left[16\hat{s}^2 (\Delta_{4F})^2 + 3\hat{c}_2 \left(4(\Delta_{4F})^2 + (c_{\Phi,1})^2 \right) + 4\Delta_{4F} c_{\Phi,1} + 16\hat{s}_2 \Delta_{4F} c_{BW} \right] \frac{\hat{v}^4}{\Lambda^4} ,$$

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Analysis Framework

- The theoretical predictions for the analyses were obtained with MadGraph5_aMC@NLO at leading order in QCD and QED, with the UFO files for the effective Lagrangian with the seventeen operators of the rotated basis generated with FeynRules including also the $\mathcal{O}(\Lambda^{-2})$ and $\mathcal{O}(\Lambda^{-4})$ terms for the renormalization of the SM inputs.
- Parton shower and hadronization was performed using PYTHIA8, and the fast detector simulation was carried out with Delphes. Jet analyses was done using FASTJET. The detector response for ATLAS NC data was simulated using Rivet

Effective combinations for DY and EWPO

- We identified eight combinations of Wilson coefficients in the *rotated basis* which can be distinguished by primarily studying the invariant mass distribution of the lepton pairs produced in Drell-Yan processes.
- They can be chosen to be the five combinations

$$\overline{\Delta}_{4F}, \overline{c}_{BW}, \overline{c}_{\Phi,1}, \overline{c}_{W^2H^4}^3, \overline{c}_{2JB}$$

together with the coefficients of the dimension-eight operators

$$c_{\psi^4H^2}^{(7)}, c_{\psi^4D^2}^{(2)}, c_{\psi^4D^2}^{(3)}$$

Modified vertices to
fermions

$$\Delta g_1^O = \frac{1}{2} \left[\hat{T} - \left(W + \frac{2}{\hat{c}^2} W' \right) + \frac{2\hat{s}}{\hat{c}} \left(X + \frac{2}{\hat{c}^2} X' \right) - \frac{\hat{s}^2}{\hat{c}^2} \left(Y + \frac{2}{\hat{c}^2} Y' \right) \right],$$

$$\Delta g_2^O = \frac{\hat{s}^2}{\hat{c}^2} \left[\hat{c}^2 \hat{T} - \hat{S} + \hat{s}^2 \left(W + \frac{1}{\hat{c}^2} W' \right) - \frac{1 - 2\hat{s}^2 \hat{c}^2}{\hat{s} \hat{c}} \left(X + \frac{1}{\hat{c}^2} X' \right) + \hat{c}^2 \left(Y + \frac{1}{\hat{c}^2} Y' \right) \right],$$

$$\Delta g_W^O = \frac{1}{2\hat{c}_2} \left[\hat{c}^2 \hat{T} - 2\hat{s}^2 \hat{S} - (1 - 3\hat{s}^2) \left(W + 2W' \right) + \hat{s}^2 \left(Y + \frac{1}{\hat{c}^2} Y' \right) - 2\hat{s} \hat{c} \left(X + \frac{1}{\hat{c}^2} X' \right) \right] - \frac{1}{2} \hat{U} + V + \frac{3}{2} V'.$$

$$\varepsilon_{\gamma\gamma}^{(\prime)} = \hat{s}^2 W^{(\prime)} + \hat{c}^2 Y^{(\prime)} + 2\hat{s} \hat{c} X^{(\prime)},$$

$$\varepsilon_{ZZ}^{(\prime)} = \hat{c}^2 W^{(\prime)} + \hat{s}^2 Y^{(\prime)} - 2\hat{s} \hat{c} X^{(\prime)},$$

$$\varepsilon_{Z\gamma}^{(\prime)} = (\hat{c}^2 - \hat{s}^2) X^{(\prime)} + \hat{s} \hat{c} (W^{(\prime)} - Y^{(\prime)}),$$

$$\varepsilon_{WW}^{(\prime)} = W^{(\prime)} - V^{(\prime)},$$

Four-fermion
contact terms

Oblique Parameters

$$\begin{aligned} \hat{S} &= \frac{\hat{c}}{\hat{s}} \frac{\hat{v}^2}{\Lambda^2} \left[b_{BW} + \frac{\hat{v}^2}{\Lambda^2} \left(b_{WBH^4}^{(1)} - \frac{g'}{4} r_{WH^4D^2}^{(1)} - \frac{g}{4} r_{BH^4D^2}^{(1)} \right) \right] \\ &= \left[\frac{\hat{c}}{\hat{s}} \bar{c}_{BW} - \frac{\hat{e}^2}{2\hat{s}^2} (\bar{c}_{2JW} + \bar{c}_{2JB}) \right] \frac{\hat{v}^2}{\Lambda^2} - \left[\frac{\hat{e}^2}{8\hat{c}\hat{s}^3} c_{\psi^4H^2}^{(7)} + \frac{\hat{e}^4}{8\hat{s}^4\hat{c}^2} (\hat{s}^2 c_{\psi^4D^2}^{(2)} + \hat{c}^2 c_{\psi^4D^2}^{(3)}) \right] \frac{\hat{v}^4}{\Lambda^4} \end{aligned}$$

$$\hat{T} = -\frac{\hat{v}^2}{2\Lambda^2} \left(b_{\Phi,1} + \frac{\hat{v}^2}{\Lambda^2} b_{H^6}^{(2)} \right) = -\frac{1}{2} \left[\bar{c}_{\Phi,1} + \frac{\hat{e}^2}{\hat{c}^2} \bar{c}_{2JB} \right] \frac{\hat{v}^2}{\Lambda^2} - \frac{\hat{e}^2}{2\hat{s}_2} c_{\psi^4H^2}^{(7)} \frac{\hat{v}^4}{\Lambda^4}$$

$$W = \frac{g^2 \hat{v}^2}{4\Lambda^2} \left[r_{2W} - \frac{\hat{v}^2}{\Lambda^2} r_{W^2H^2D^2}^{(9)} \right] = -\frac{\hat{e}^2}{2\hat{s}^2} \bar{c}_{2JW} \frac{\hat{v}^2}{\Lambda^2} = \bar{\Delta}_{4F} \frac{\hat{v}^2}{\Lambda^2}$$

$$Y = \frac{g^2 \hat{v}^2}{4\Lambda^2} \left[r_{2B} - \frac{\hat{v}^2}{\Lambda^2} r_{B^2H^2D^2}^{(9)} \right] = -\frac{\hat{e}^2}{2\hat{s}^2} \bar{c}_{2JB} \frac{\hat{v}^2}{\Lambda^2}$$

$$\hat{U} = \frac{\hat{v}^4}{\Lambda^4} \left[b_{W^2H^4}^{(3)} + \frac{g'}{2} r_{WH^4D^2}^{(2)} \right] = \left[\bar{c}_{W^2H^4}^{(3)} - \frac{\hat{e}^2}{2\hat{s}_2} c_{\psi^4H^2}^{(7)} - \frac{\hat{e}^4}{2\hat{s}_2^2} c_{\psi^4D^2}^{(2)} \right] \frac{\hat{v}^4}{\Lambda^4}$$

$$X = \frac{g^2 \hat{v}^4}{8\Lambda^4} r_{BWH^2D^2}^{(13)} = \frac{\hat{e}^2}{8\hat{s}^2} c_{\psi^4H^2}^{(7)} \frac{\hat{v}^4}{\Lambda^4}$$

$$W' = -\frac{g^4 \hat{v}^4}{8\Lambda^4} r_{W^2D^4}^{(1)} = -\frac{\hat{e}^4}{8\hat{s}^4} c_{\psi^4D^2}^{(3)} \frac{\hat{v}^4}{\Lambda^4}$$

$$Y' = -\frac{g^4 \hat{v}^4}{8\Lambda^4} r_{B^2D^4}^{(1)} = -\frac{\hat{e}^4}{8\hat{s}^4} c_{\psi^4D^2}^{(2)} \frac{\hat{v}^4}{\Lambda^4}$$

$$X' = V = V' = 0$$